The Fokker–Planck Equation

Gardiner, *Stochastic Methods*, Chapter 5

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Interdisciplinary Journal Club

October 28, 2026



Outline

1. The Fokker–Planck equation: where it comes from
2. The probability current
3. Boundary conditions and topology
4. Eigenfunction methods
5. Stationary and quasi-stationary states
6. First passage
7. Examples

The Differential Chapman–Kolmogorov Equation

Drift, diffusion, and jumps

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- Also called the forward Kolmogorov (or Smoluchowski) equation

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- Forward: fix the start, evolve the end. Backward: fix the end, vary the start
- Natural for questions about the start: survival, exit times and splitting probabilities are all such averages

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  - With $B = D$: $\langle (\Delta x)^2 \rangle = Dt$. Many texts write $\partial_t p = D'\partial_x^2 p$, with $D' = D/2$

A General Physical Model

Damped Brownian particle with inertia

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  - $\tau_x = \gamma/|\mathcal{U}''|$: the position slides down a well of curvature $\mathcal{U}''$

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  - So $m\,\Delta v$ is negligible: friction balances force plus noise. Set $m\,\mathrm{d}v \approx 0$:

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- Valid for $t \gg \tau_v$, with corrections of order ε

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  - The first correction multiplies the diffusion by $1 + m\mathcal{U}''/\gamma^2 = 1 \pm \varepsilon$

Probability Current

Definition and local conservation

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- In arbitrary dimensions:

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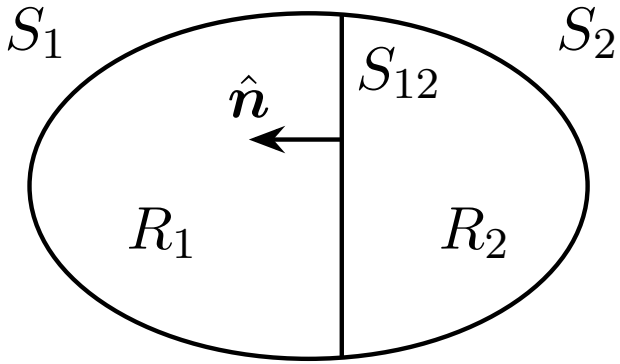
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- Probability leaves R only through its surface, like charge

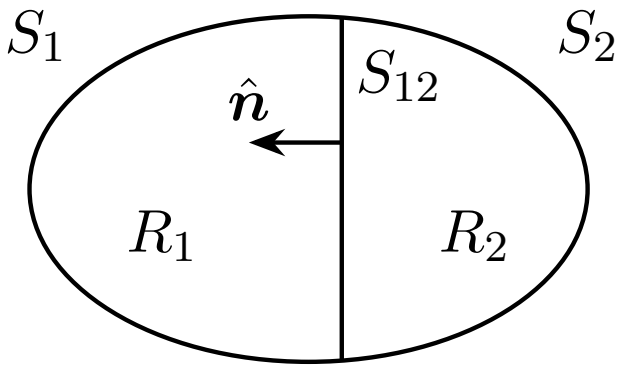
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Flux across any surface



Probability Current

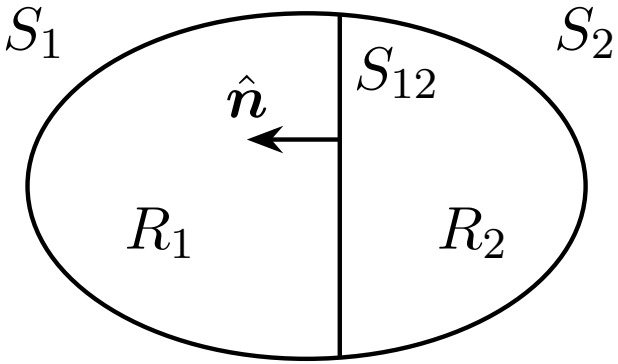
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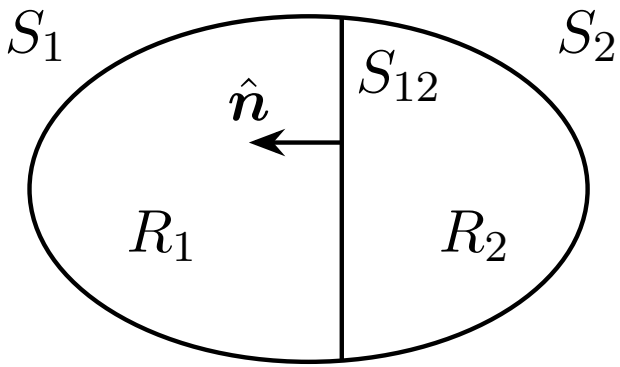


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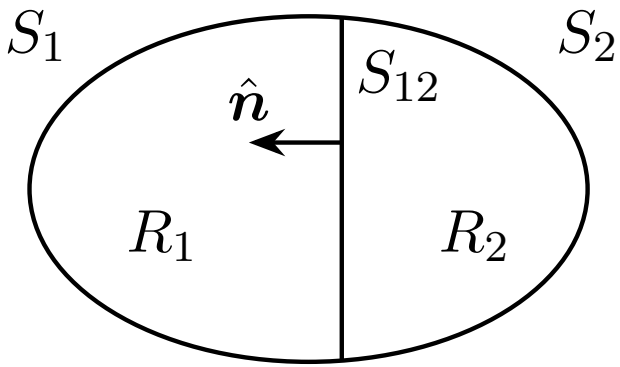
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- The continuity equation alone fixes $\boldsymbol{J}$ only up to a divergence-free field; this shows that the $\boldsymbol{J}$ above is the physical flux

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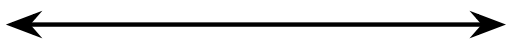
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  - Either zero everywhere, or a steady current around a loop

Boundary Conditions

Four domains in one dimension

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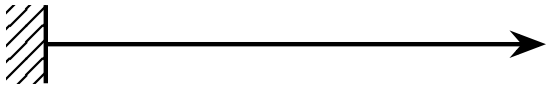
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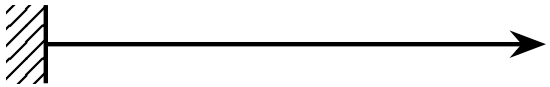
half-line $(0, \infty)$: one wall

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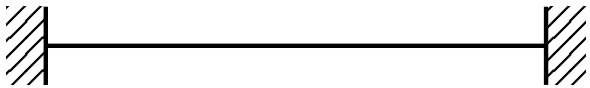
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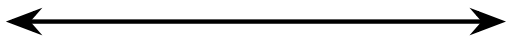
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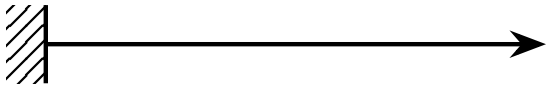
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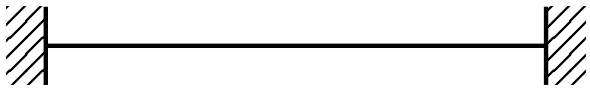
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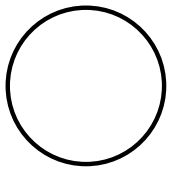
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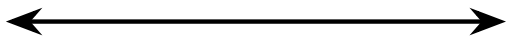
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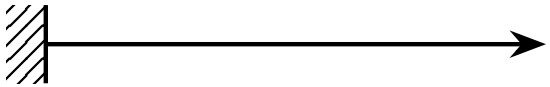
circle $S^1 = \mathbb{R}/L\mathbb{Z}$: no boundary at all

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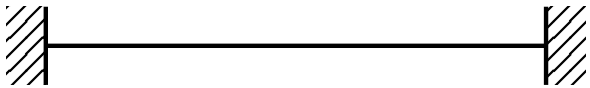
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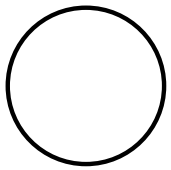
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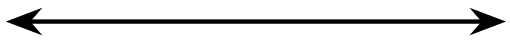


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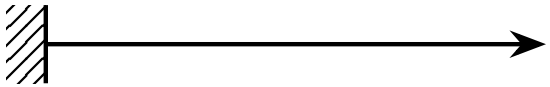
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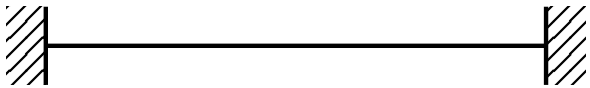
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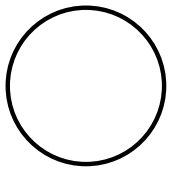
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- Boundary conditions are part of the definition of the problem
- The same differential operator with different walls is a different operator, with a different spectrum

Boundary Conditions

Reflecting and absorbing walls

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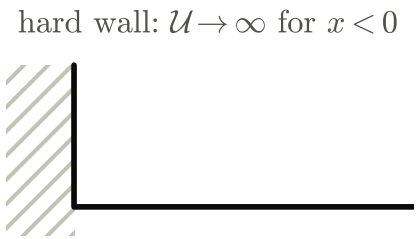
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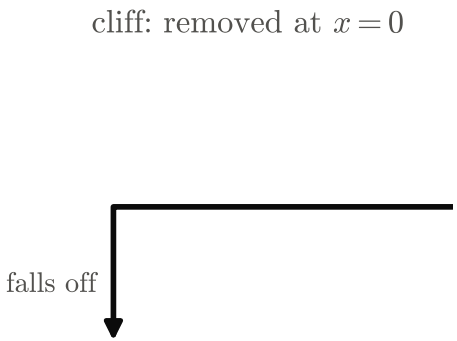
- At an interface where A or B jumps: $\hat{n}_i \, J^i$ and p are continuous

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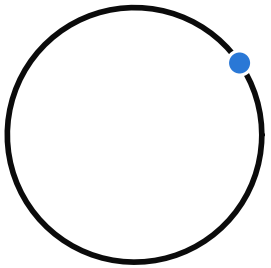
Reflecting, absorbing, and periodic



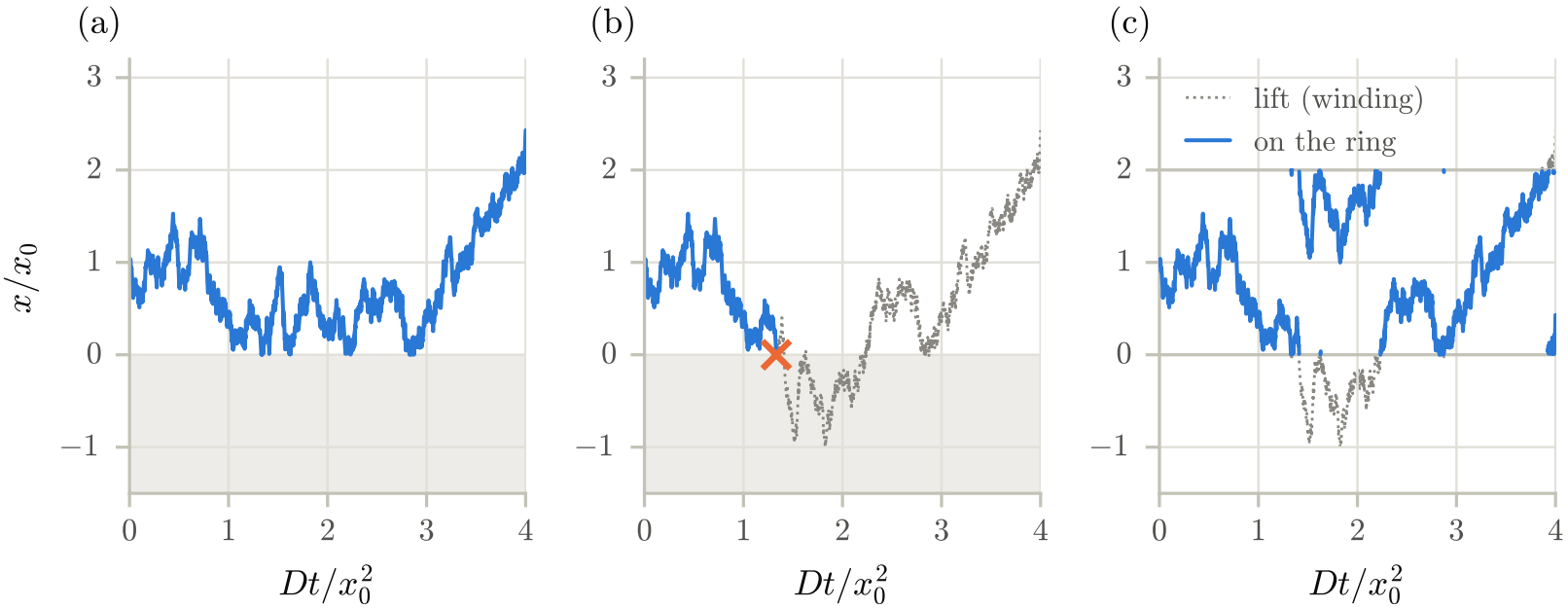
hard wall: $\mathcal{U} \to \infty$ for $x < 0$



cliff: removed at $x = 0$



ring of circumference $L = 2x_0$



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- In general (Feller) there are four types: these three, plus **regular**, which is where we choose

Feller's Classification

The general test

Feller's Classification

The general test

- With x_0 inside the domain, define

$$f(x) = \exp\Big[-2\int_{x_0}^{x} \frac{A}{B}\,\mathrm{d}s\Big], \qquad g(x) = \frac{2}{B(x)\,f(x)},$$

$$h_1(x) = f(x)\int_{x_0}^{x} g\,\mathrm{d}s, \qquad h_2(x) = g(x)\int_{x_0}^{x} f\,\mathrm{d}s$$

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- Integrability on (a, x_0) decides the type of the boundary a:

Type	On (a, x_0)	Reached?	Left?	Condition at a
Regular	f, g integrable	yes	yes	must be chosen
Exit	g not integrable, h_1 integrable	yes	no	none (absorbs)
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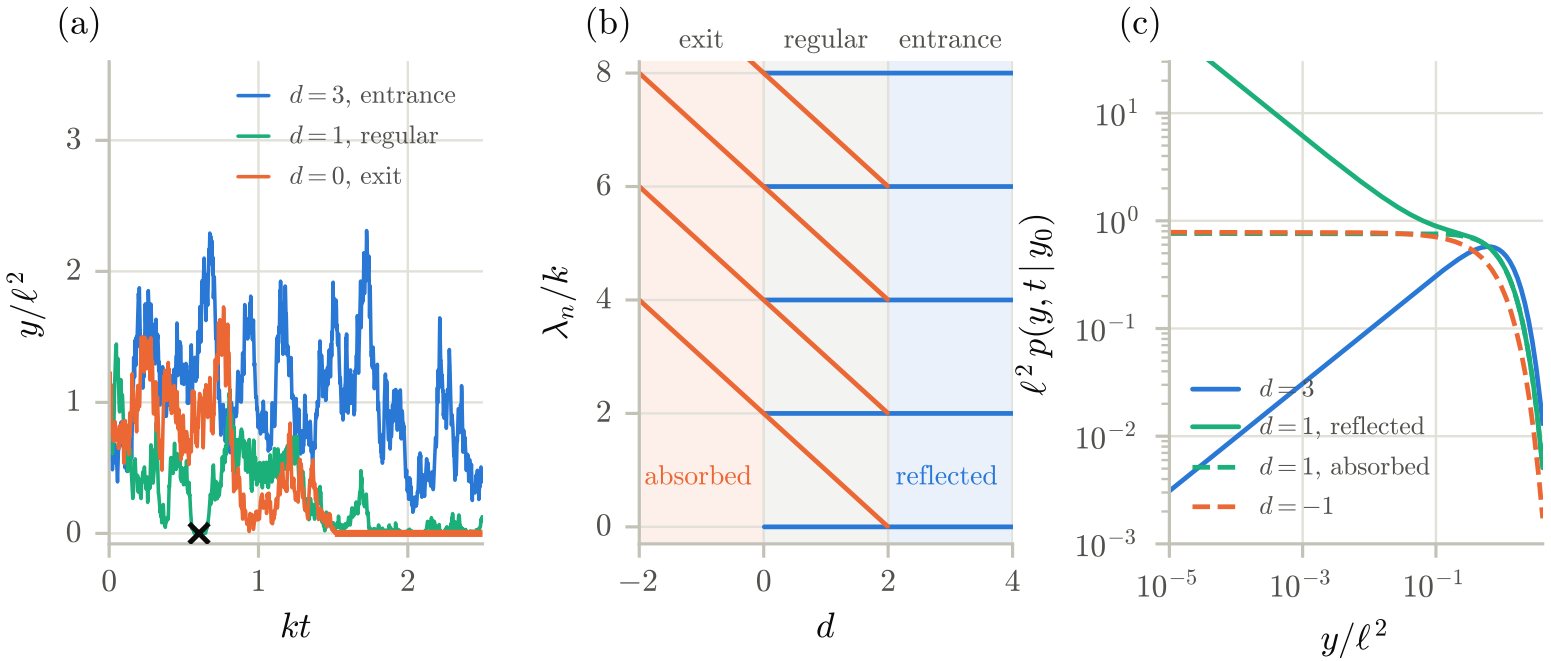
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- Exit: reached in finite mean time; no normalisable p_s.
  Entrance: p_s can exist; never reached

Feller's Classification

Plot Lab

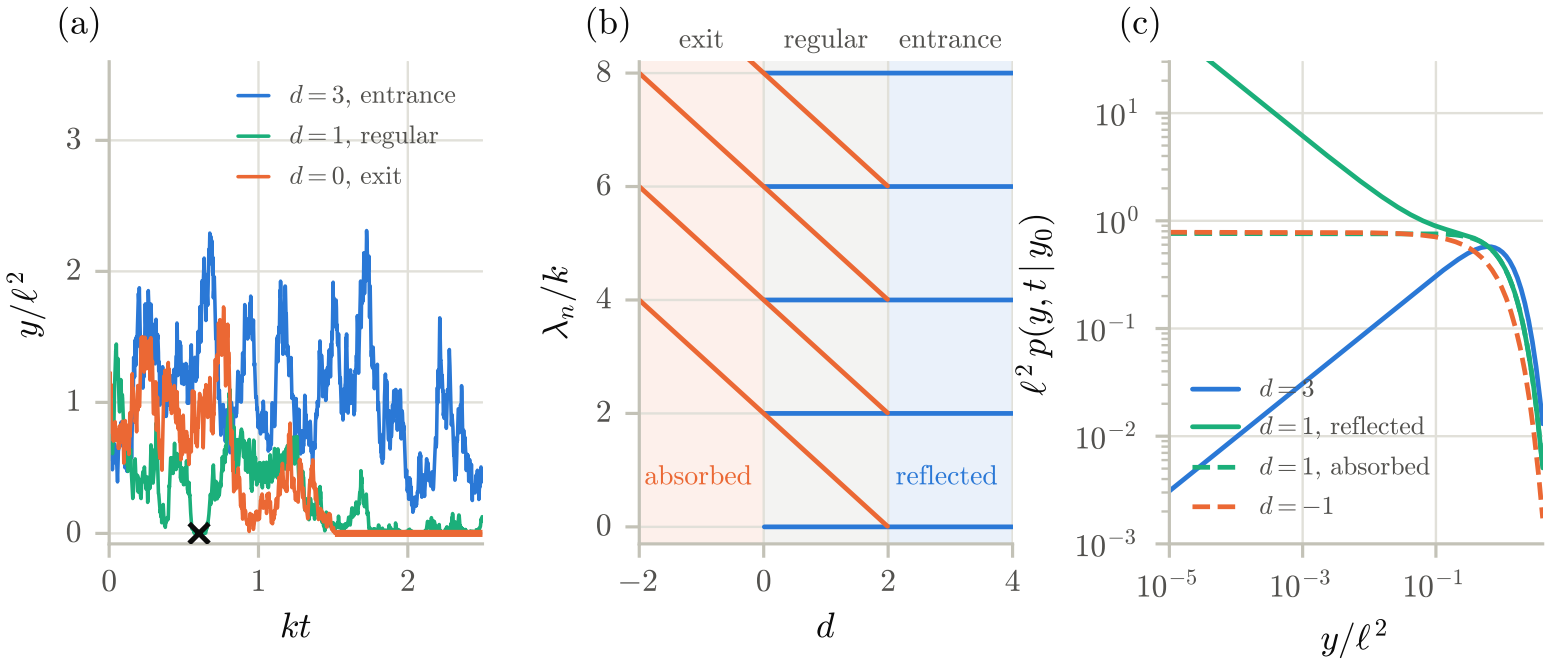
Example: the squared radius



Feller's Classification

▸ Plot Lab

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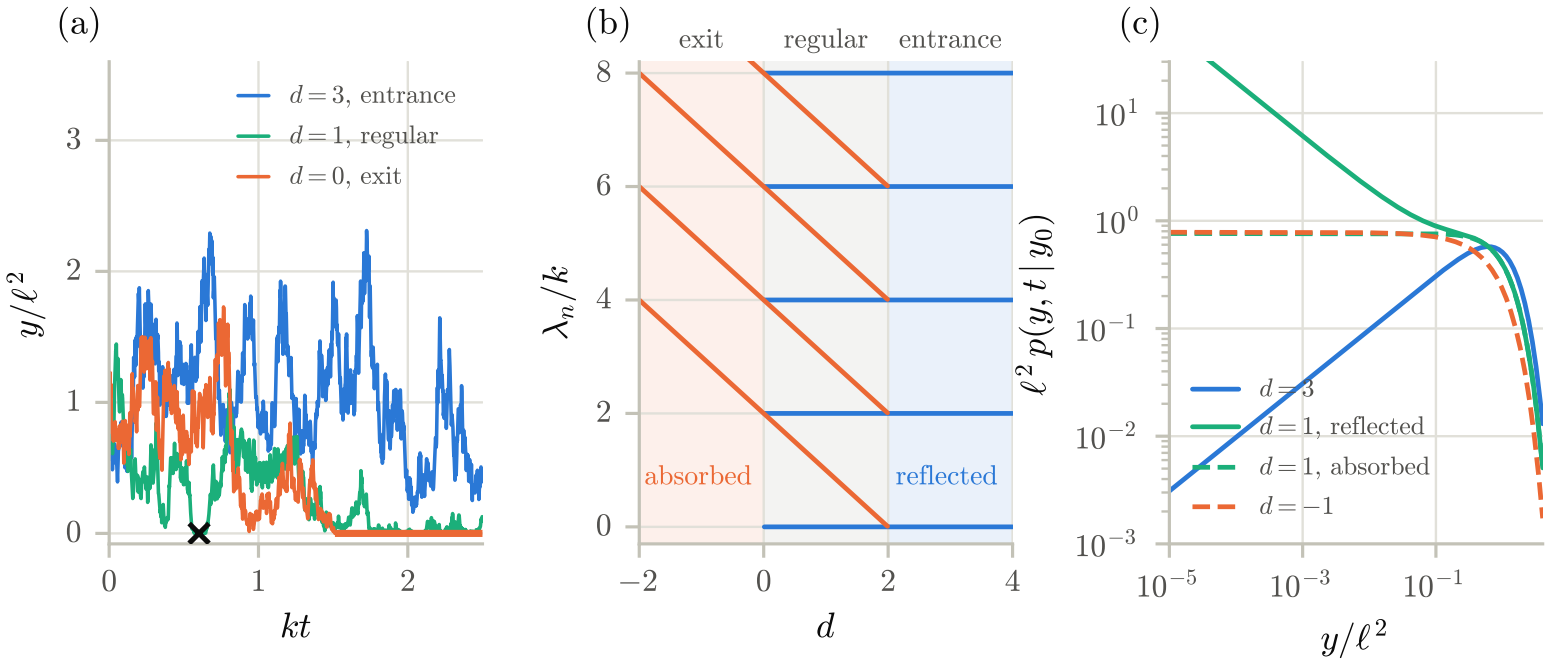


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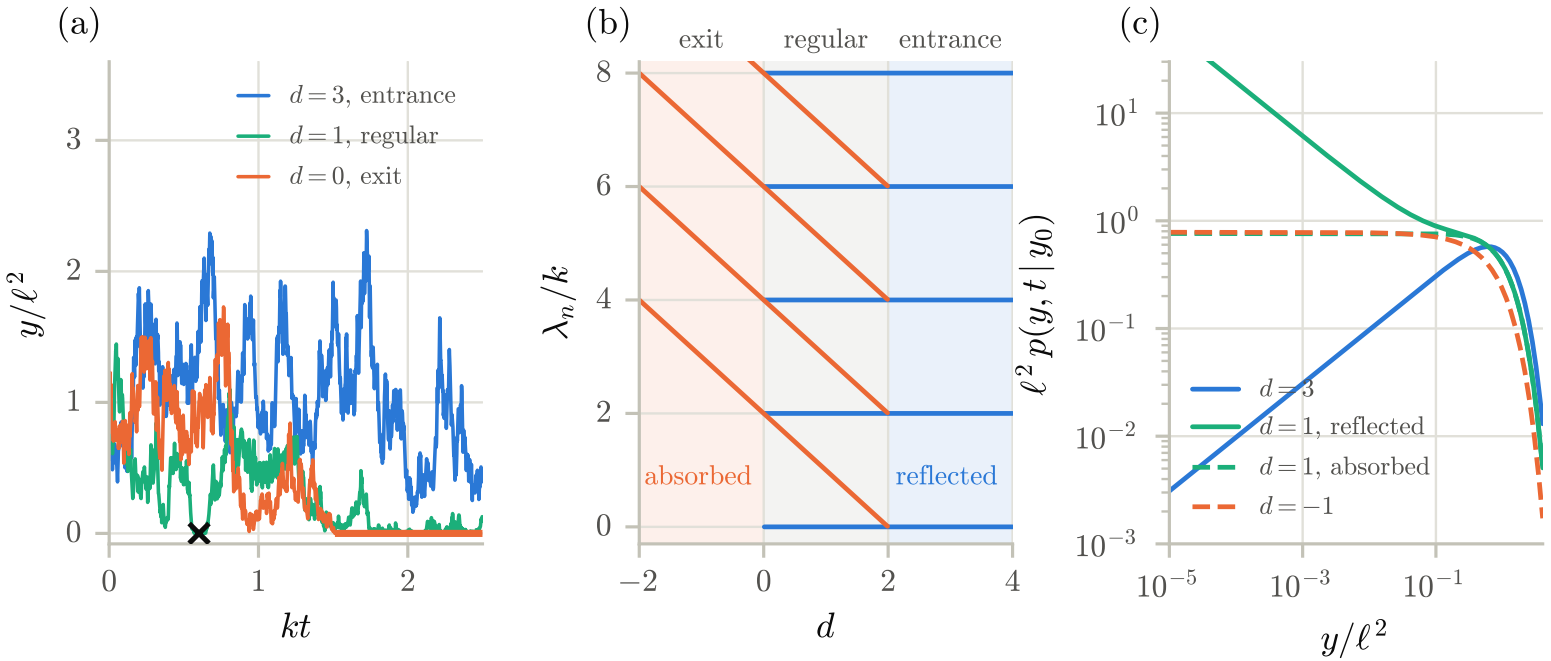


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- The quick rule fails here ($\sqrt{B}$ not Lipschitz): $A(0) = dD > 0$ for all $d > 0$

It's Linear Algebra Time

The forward and backward operators

It's Linear Algebra Time

The forward and backward operators

- Homogeneous process: p depends on $t - t'$ only, so take $t' = 0$. Forward (Fokker–Planck) operator, acting on the final point x:

$$\partial_t\, p(x, t \,|\, y, 0) = \mathcal{L}_x\, p(x, t \,|\, y, 0), \quad \mathcal{L}p = -\partial_x(Ap) + \tfrac{1}{2}\,\partial_x^2(Bp)$$

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  - $\mathcal{L}$ moves probability forward in time; $\mathcal{L}^\dagger$ (the *generator*) moves observables

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Adjoint operators

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- Densities p ($\int |p| < \infty$: the space L^1) and bounded observables f (the space L^∞) are paired by the expectation value, $\langle f, p \rangle = \int_a^b f\, p \, \mathrm{d}x$

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- In $L^2(1/p_s)$, $\langle p, q \rangle = \int pq/p_s \, \mathrm{d}x$: $\mathcal{L}$ *is* symmetric, the P_λ orthonormal

Eigenfunction Methods

Expanding any solution

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Imaginary time

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- Differences: $\lambda \geq 0$ are decay rates; p itself (not $|\psi|^2$) is the probability; $Q_\lambda \neq P_\lambda^*$

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- Examples (constant $B(x) = D$):

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From the eigenfunction expansion

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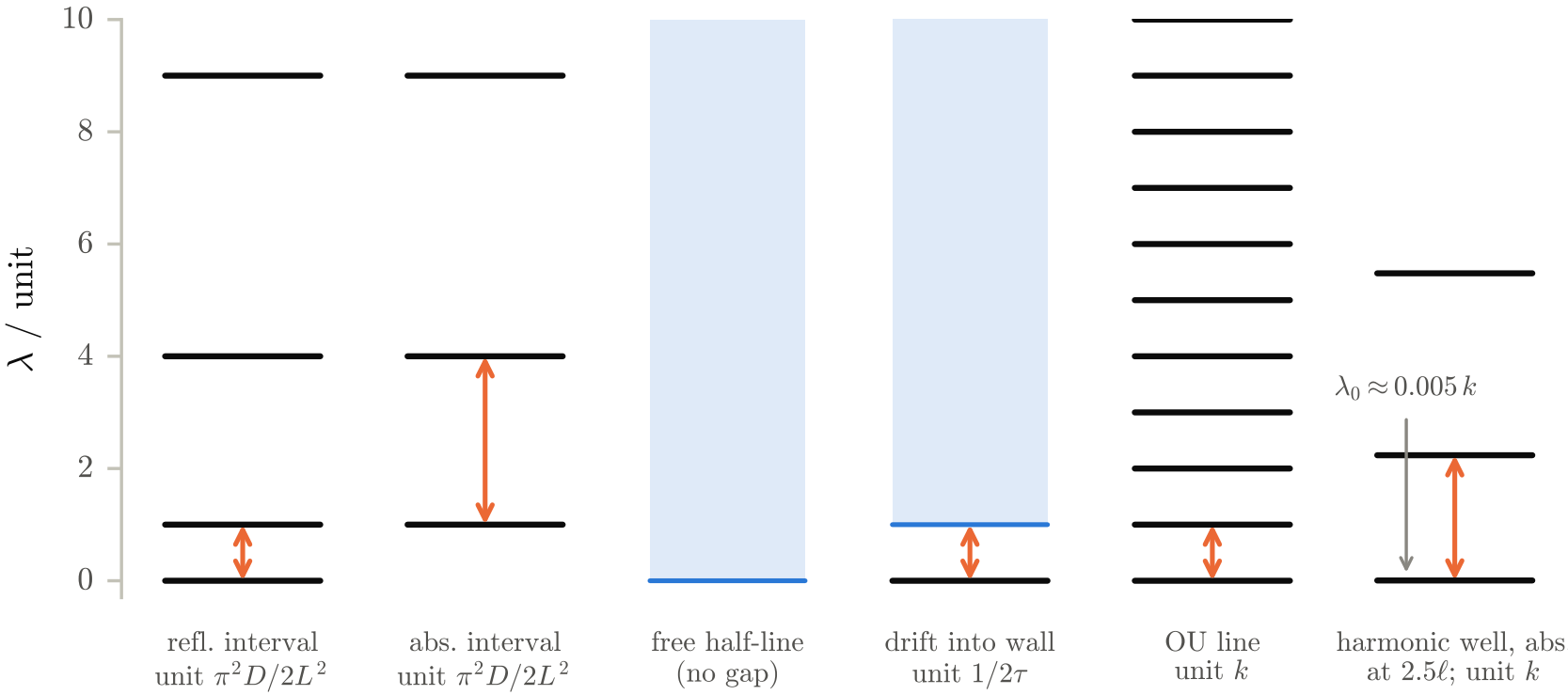
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  - The other modes only move probability around, so each changes sign

The Spectral Gap

▸ Plot Lab

Example spectra



Quasi-Stationary Distributions

Absorbing walls

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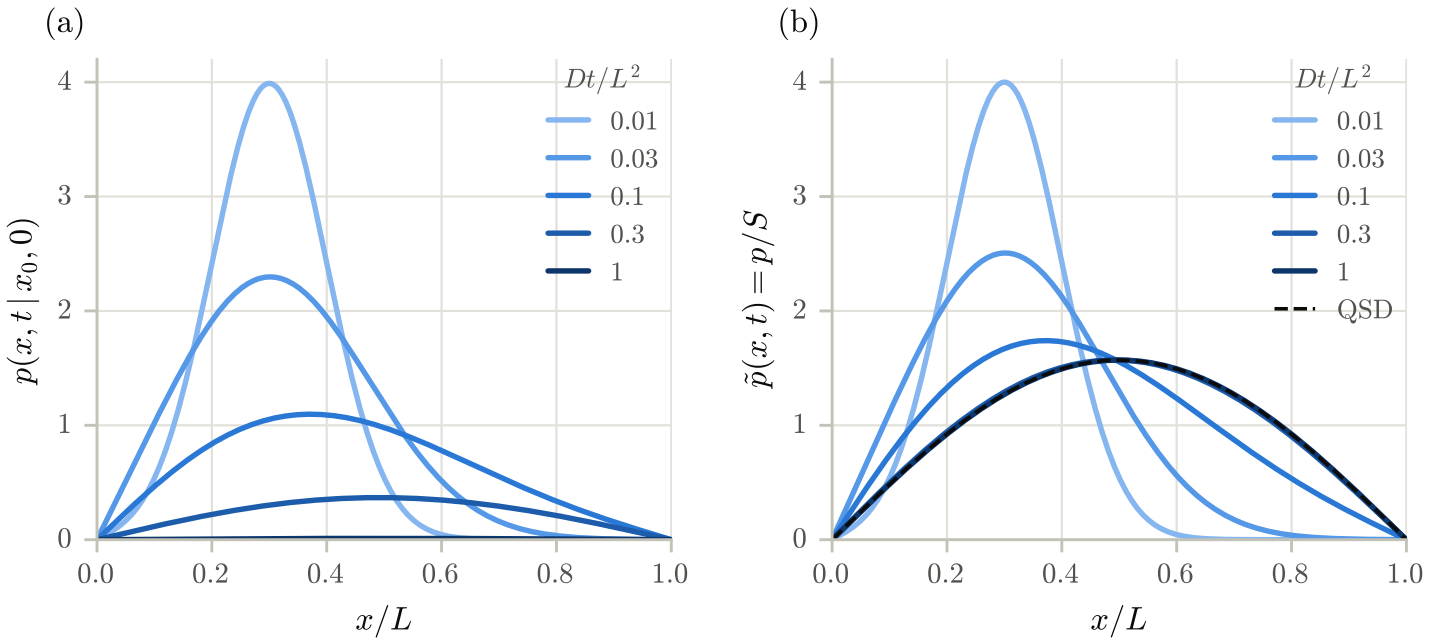
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- Two rates: λ_1 is the absorption rate; $\lambda_2 - \lambda_1$ is the rate of approach to the QSD
- Metastability: if $\lambda_2 - \lambda_1 \gg \lambda_1$, the particle equilibrates locally long before it escapes, and escape is Poisson at rate λ_1

Quasi-Stationary Distributions

▸ Plot Lab ▸ SDE Explorer

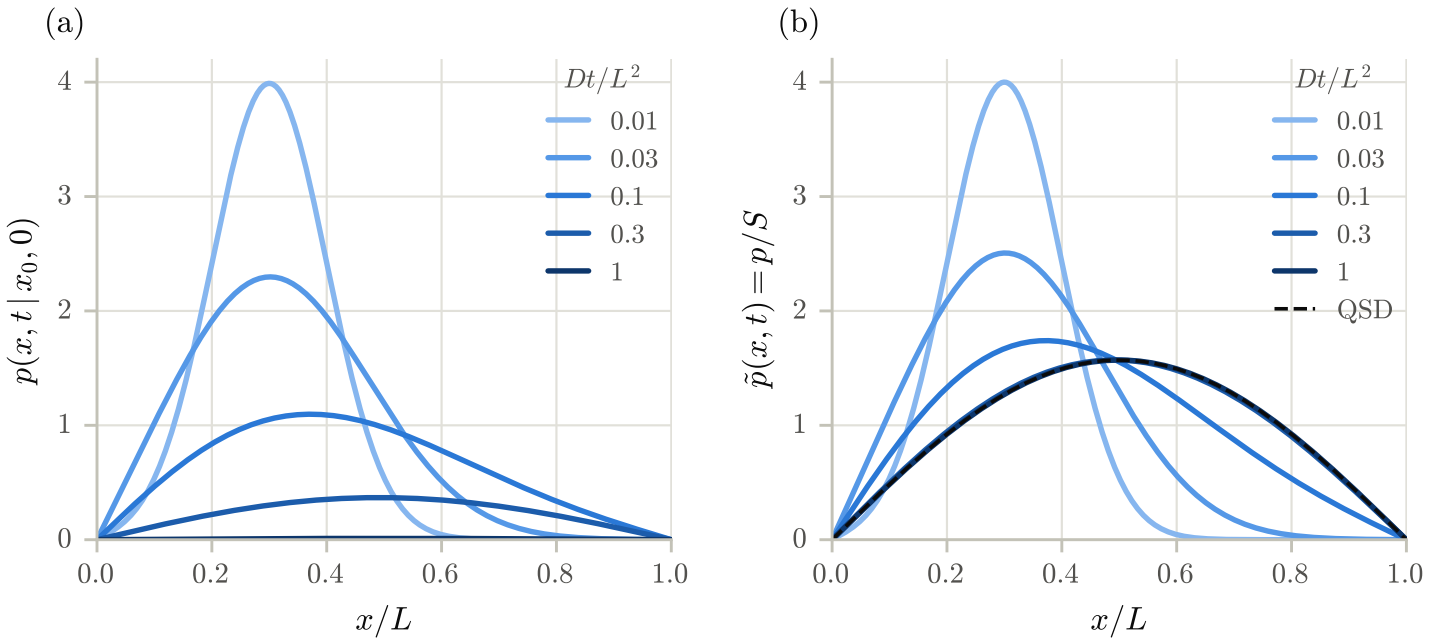
Example: free diffusion between two cliffs



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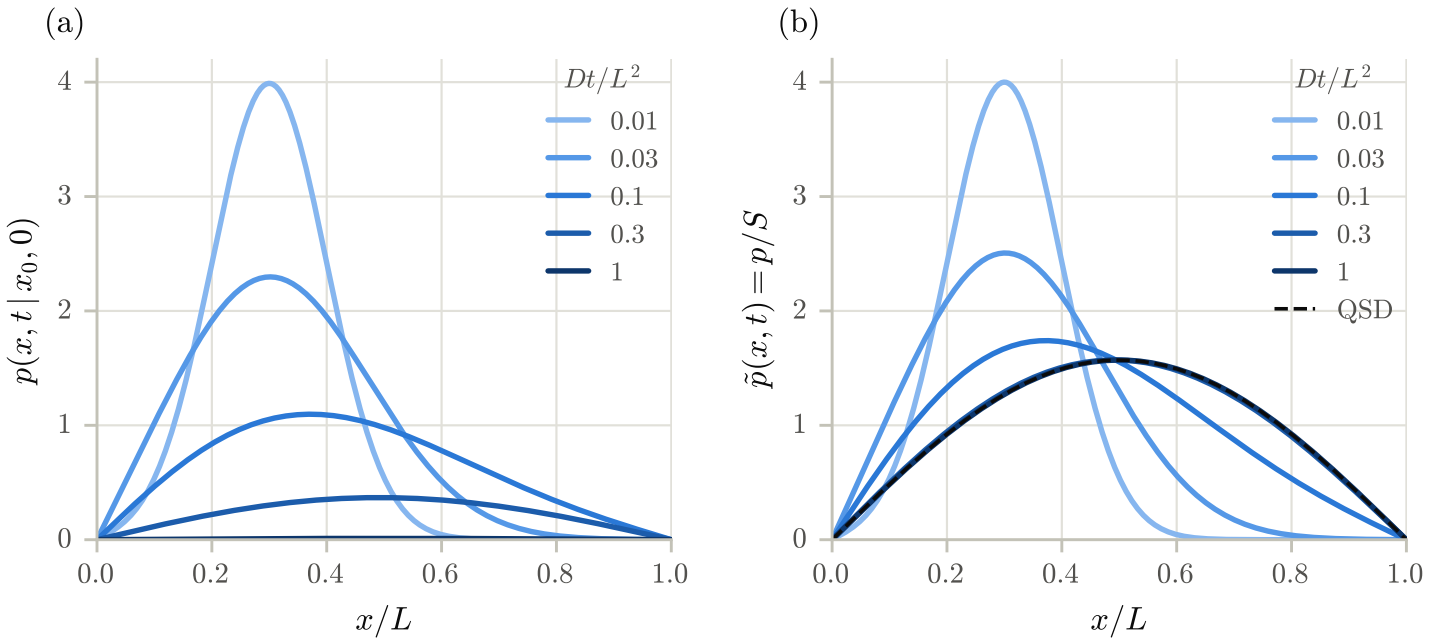


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- (a) The density shrinks; (b) $\tilde{p} = p/S$ settles on the QSD $\frac{\pi}{2L}\sin(\pi x/L)$, at rate $\lambda_2 - \lambda_1 = 3\lambda_1$

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- With a continuum, the initial distribution selects the λ that dominates at long times
- Unique, attracting QSD iff ∞ is an *entrance* boundary: the mean absorption time stays bounded as $x_0 \to \infty$

First Passage

Survival probability

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- Absorbing walls at a and b, start at $x_0 \in (a, b)$, exit time T:

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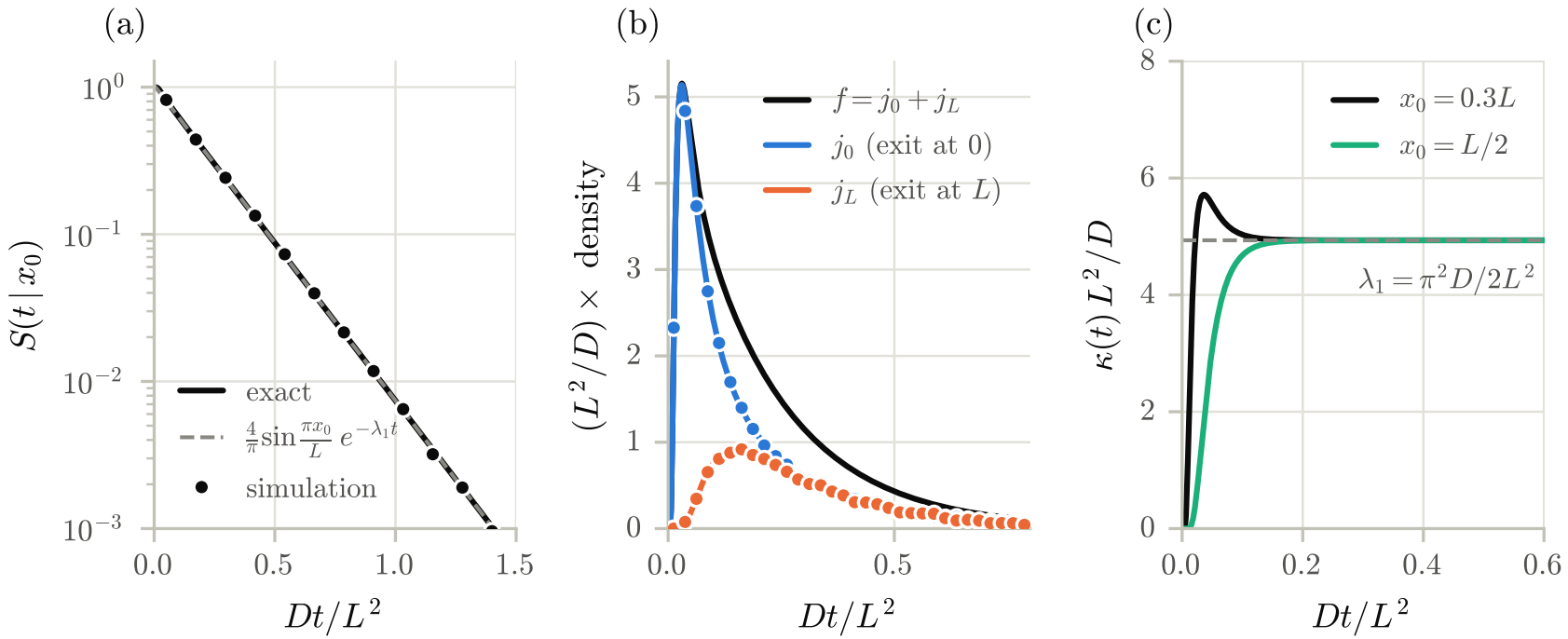
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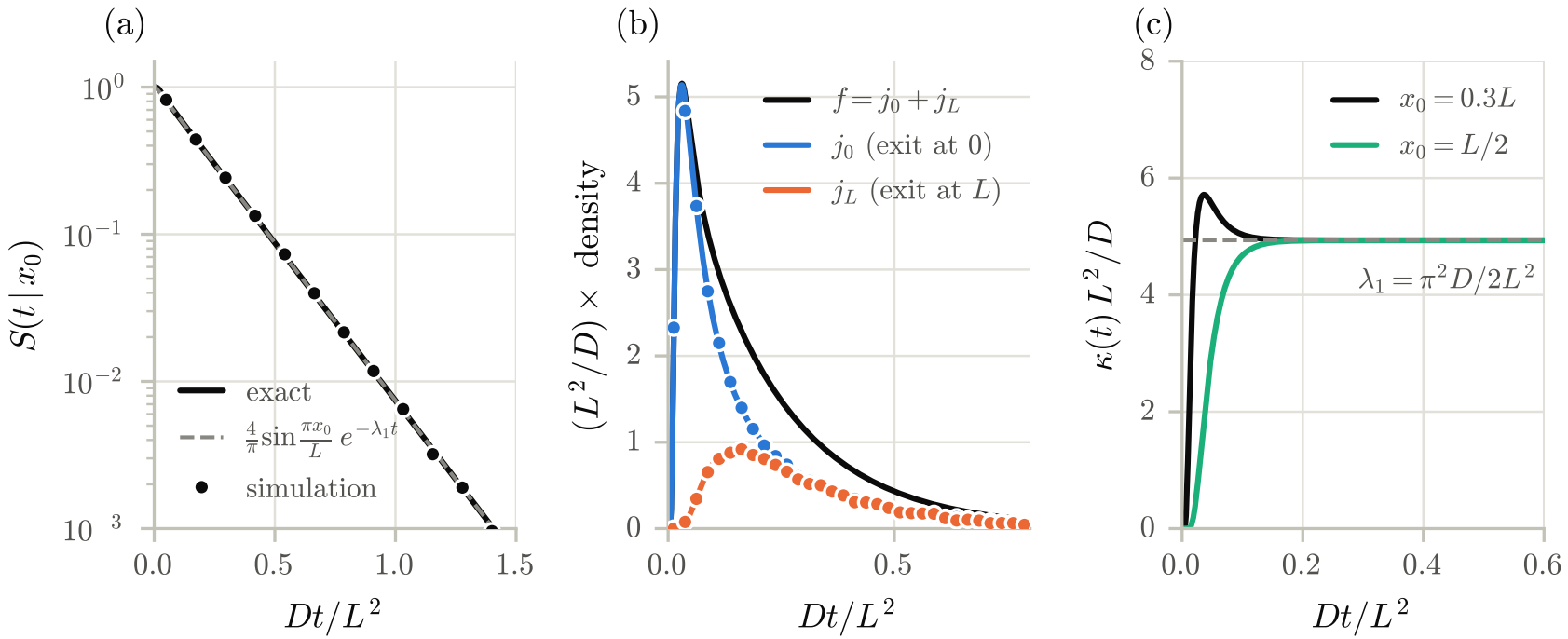
First Passage

Example: free diffusion between two cliffs



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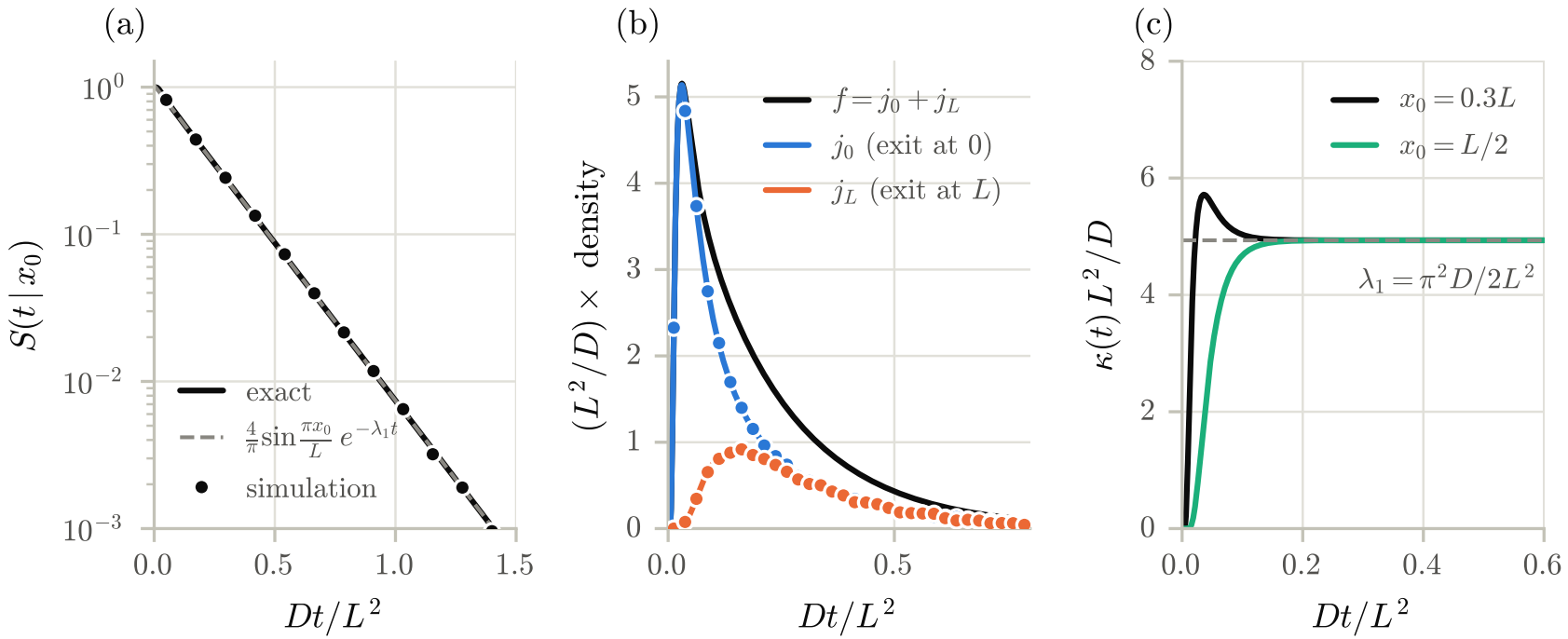
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- (a) Survival: exact, one-mode approximation, and simulation

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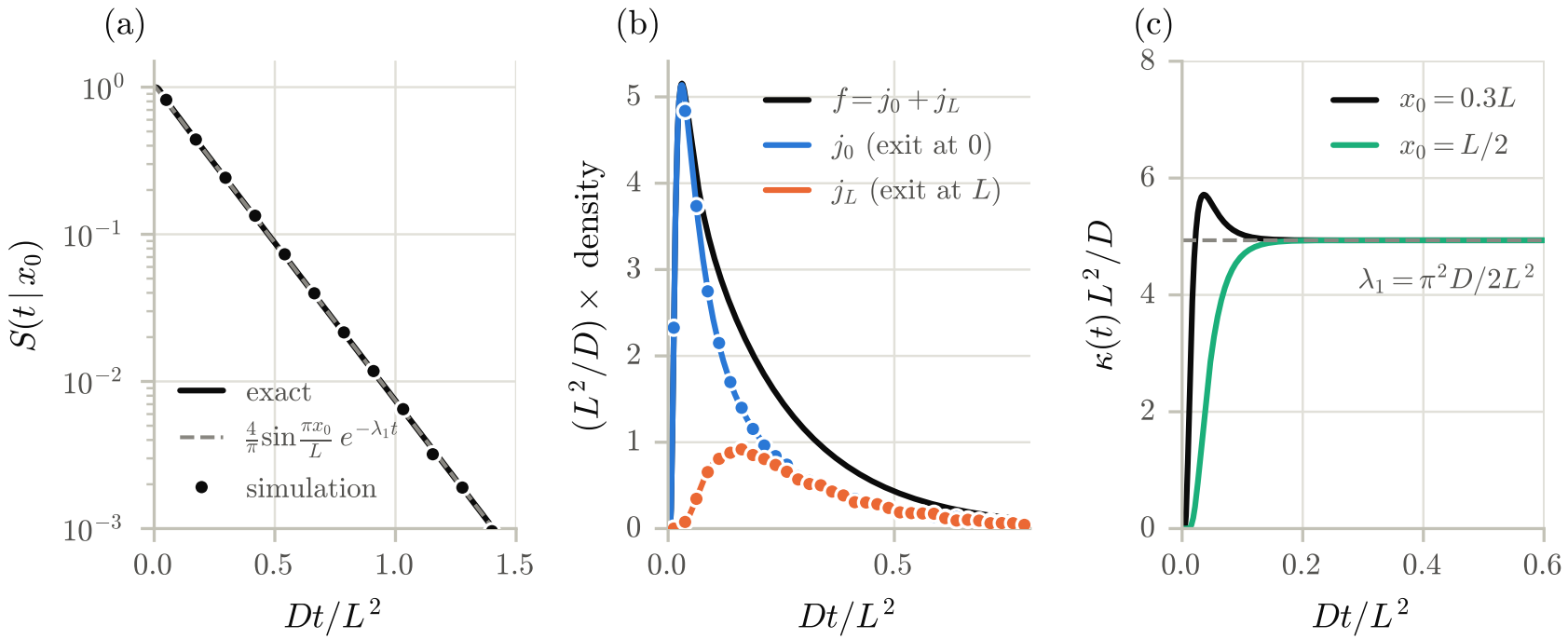
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- (a) Survival: exact, one-mode approximation, and simulation
- (b) Exit-time density, split into the currents out through each wall
- (c) Exit rate $-\dot{S}/S$ tends to $\lambda_1 = \pi^2 D/2L^2$ from every start

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- Free diffusion, both walls absorbing: $T(x_0) = (x_0 - a)(b - x_0)/D$

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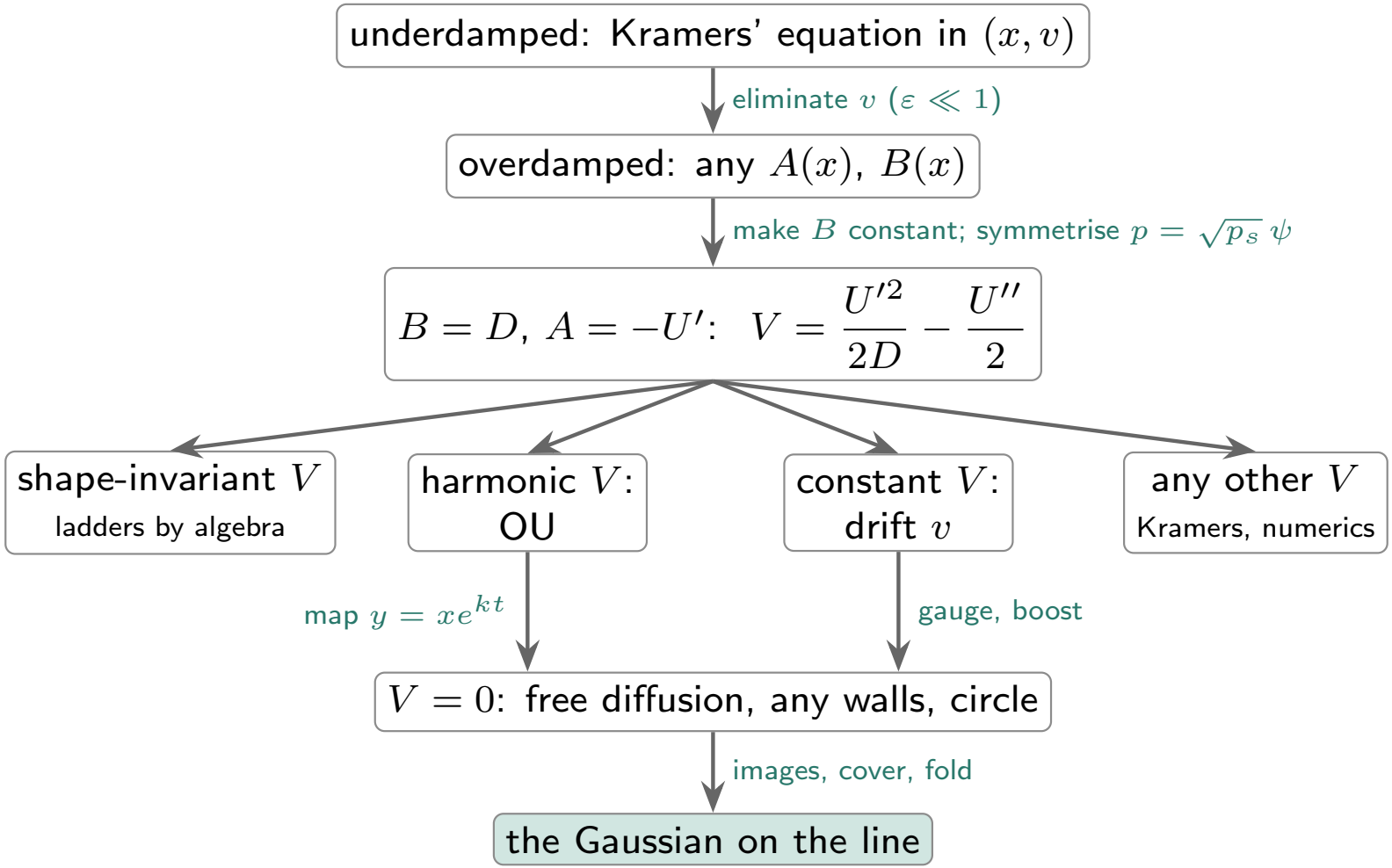
- Free diffusion: $\pi_b = (x_0 - a)/(b - a)$
- Constant drift v:

$$\pi_b = \frac{1 - e^{-2v(x_0-a)/D}}{1 - e^{-2v(b-a)/D}}$$

Examples

▸ Plot Lab

How the problems reduce to each other



Example: Free Diffusion on an Interval

One operator, two kinds of wall

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One operator, two kinds of wall

- $A = 0$, $B = D$ on $(0, L)$. Images: mirror the line's Gaussian g in both walls, then the mirrors in each other:

$$p(x, t \mid x_0, 0) = \sum_{m \in \mathbb{Z}} \left[g(x - x_0 + 2mL, t) \pm g(x + x_0 + 2mL, t) \right]$$

Example: Free Diffusion on an Interval

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Long times	$p_s = 1/L$	QSD $\propto \sin(\pi x/L)$
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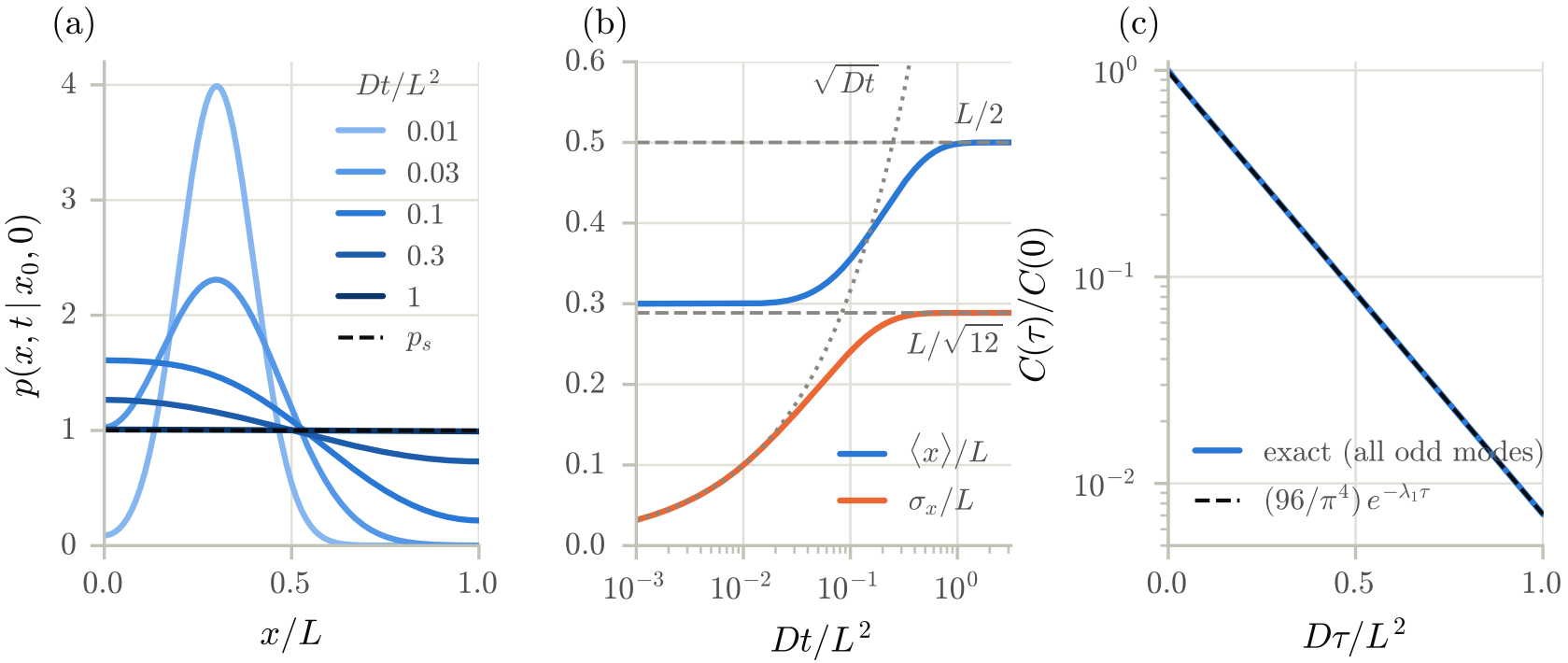
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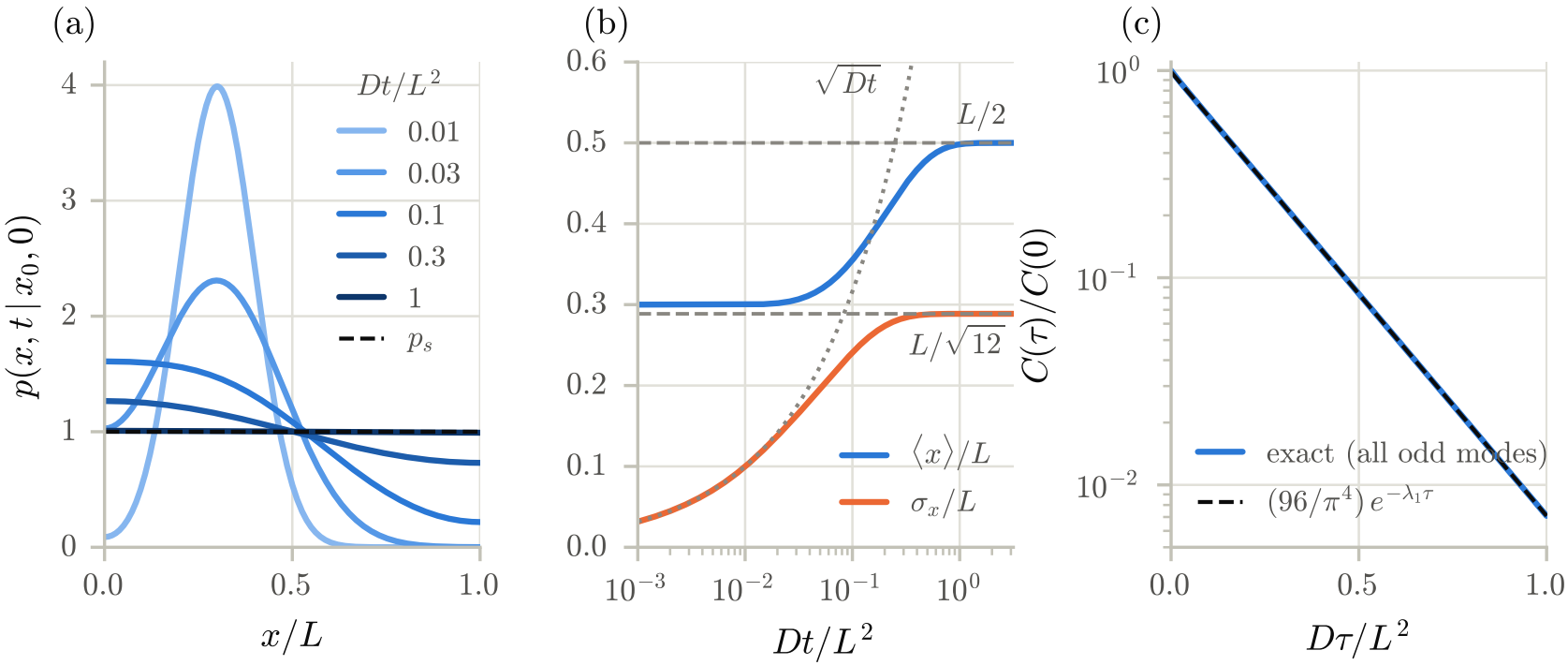
Example: Free Diffusion in a Box

▸ Plot Lab ▸ SDE Explorer



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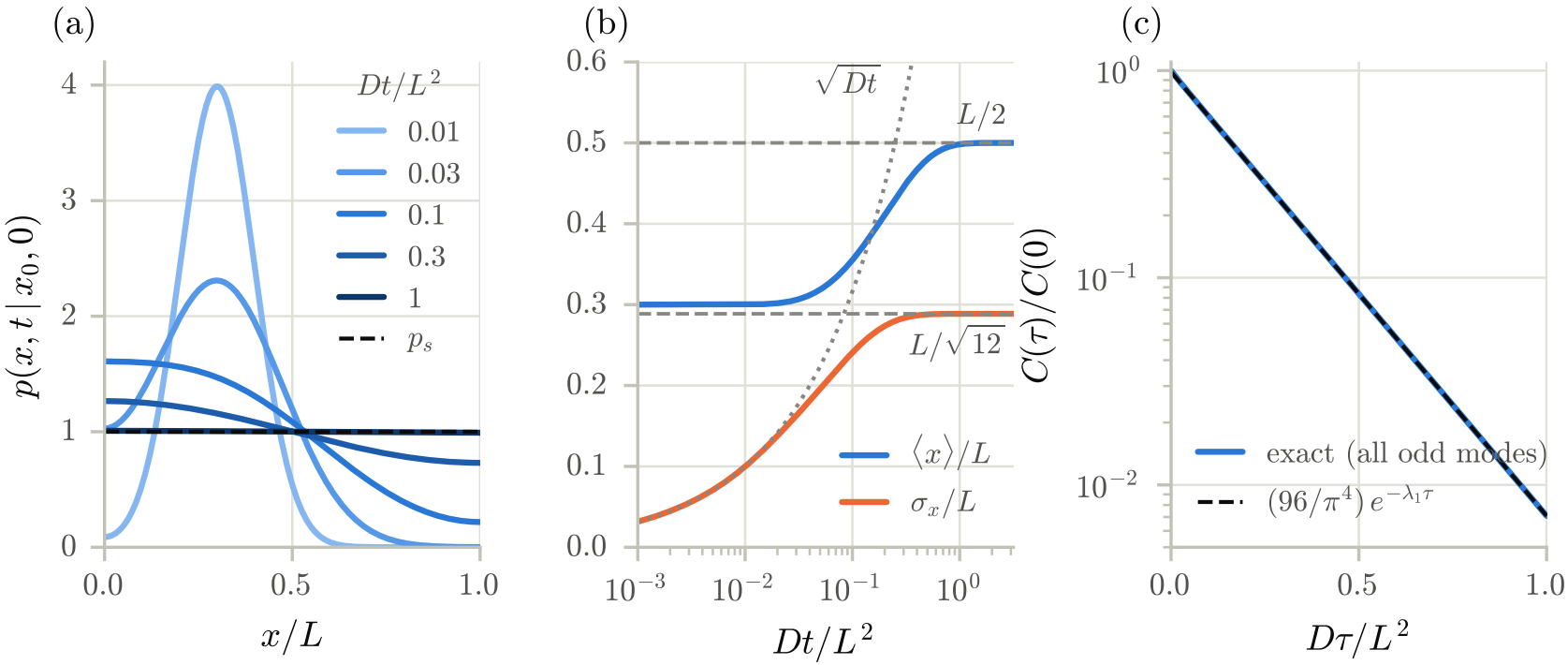
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- (a) From $x_0 = 0.3L$: a spreading Gaussian, then uniform

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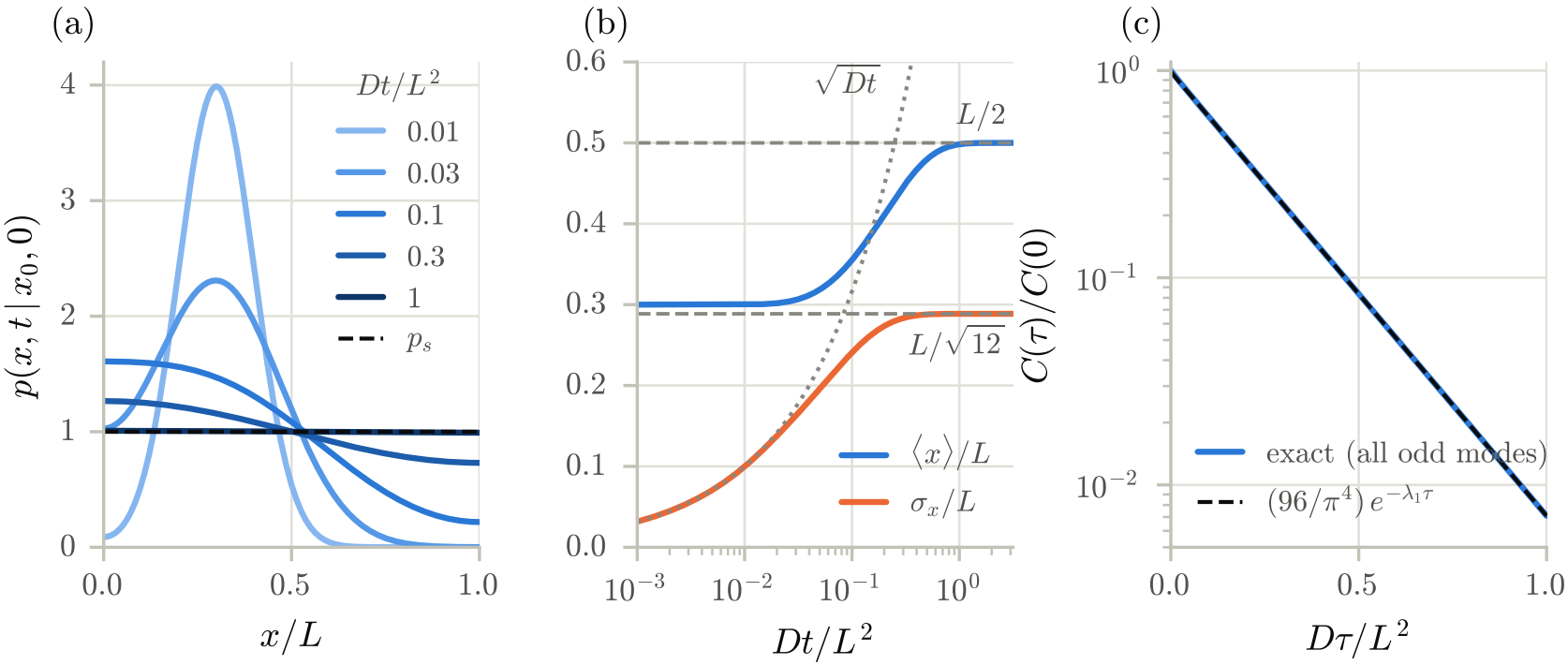
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▸ Plot Lab ▸ SDE Explorer



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- (b) σ_x grows like $\sqrt{Dt}$, then saturates at $L/\sqrt{12}$
- (c) $C(\tau)/C(0) \simeq (96/\pi^4)\,e^{-\lambda_1\tau}$: the gap dominates from the start

Example: Constant Drift on an Interval

Reduction to free diffusion

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- $A = v$, $B = D$ on $(0, L)$: sedimentation, an ion driven across a membrane. Scales $\ell = D/|v|$, $\tau = D/v^2$; one parameter, the Péclet number $vL/D = \pm L/\ell$

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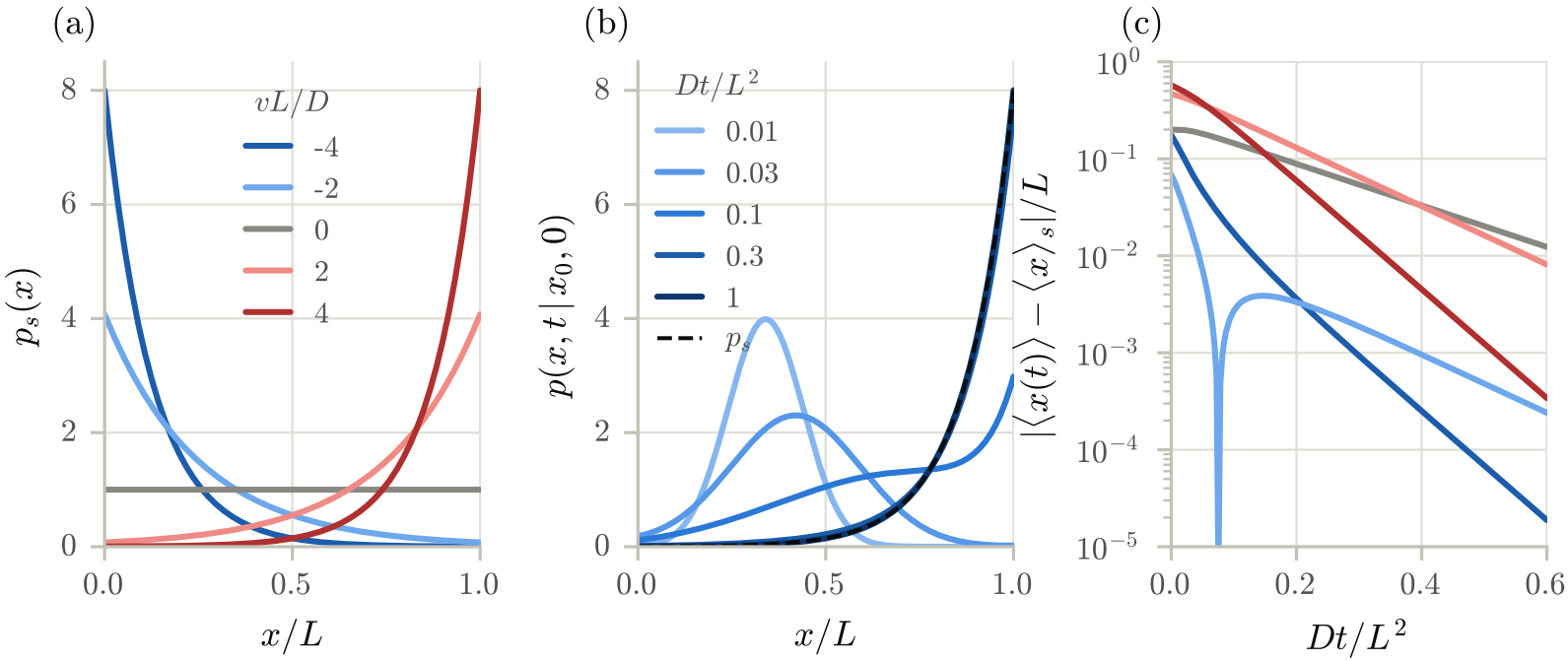
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Walls at 0, L	Picture	Rates
reflecting, reflecting	closed column	$0;\ \ 1/2\tau + n^2\pi^2 D/2L^2,\ n \geq 1$
absorbing, absorbing	two cliffs	$1/2\tau + n^2\pi^2 D/2L^2,\ n \geq 1$
reflecting, absorbing	floor and rim	$1/2\tau + Dk^2/2,\ \ \tan kL = -k/\alpha$

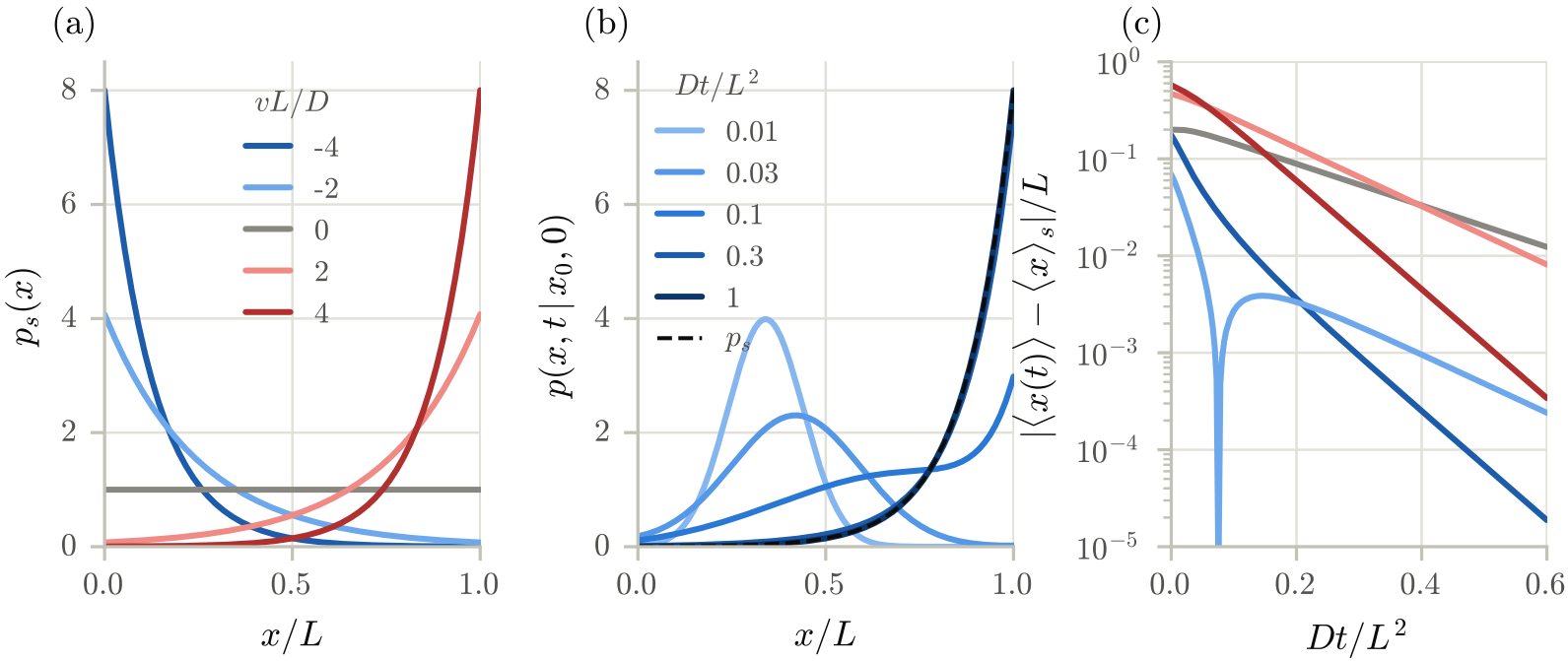
Constant Drift: Reflecting Walls

▸ Plot Lab ▸ SDE Explorer



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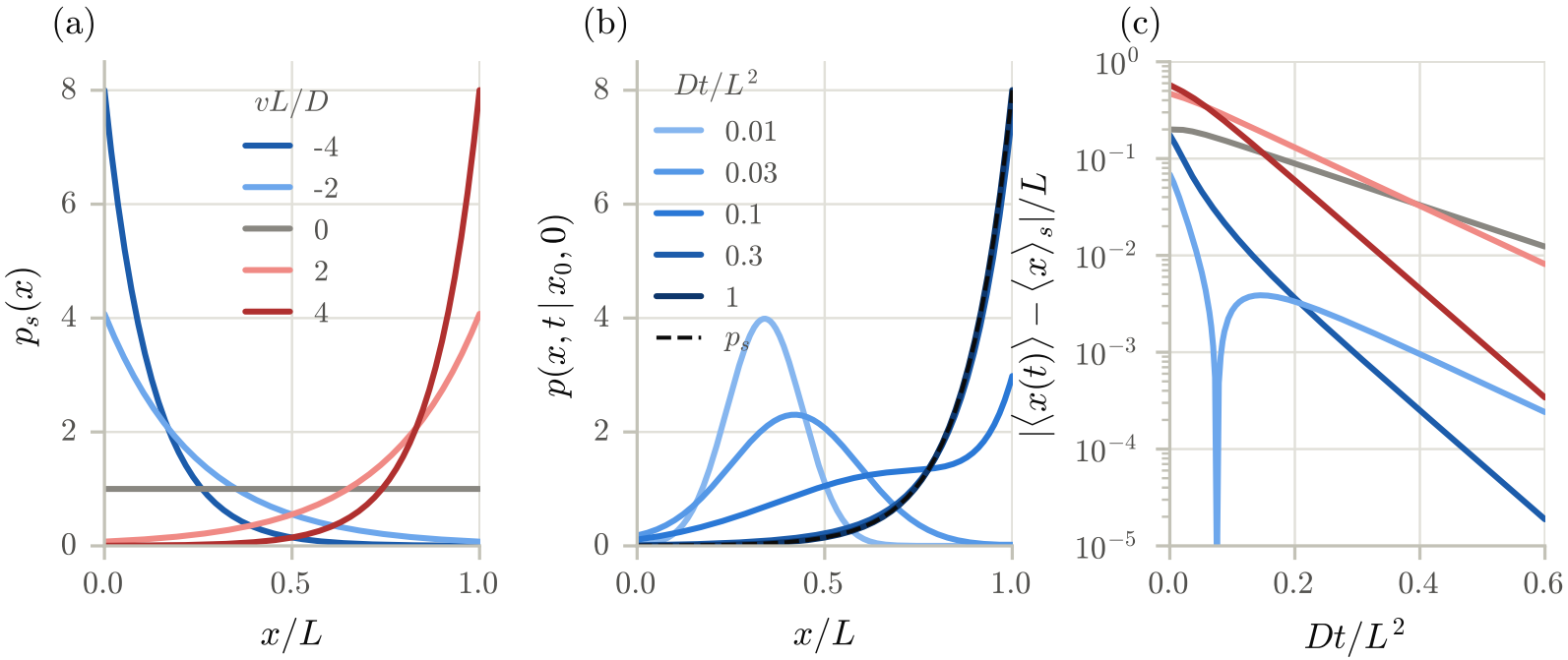
▸ Plot Lab ▸ SDE Explorer



- (a) $p_s \propto e^{2vx/D}$: barometric, piled against the downstream wall

Constant Drift: Reflecting Walls

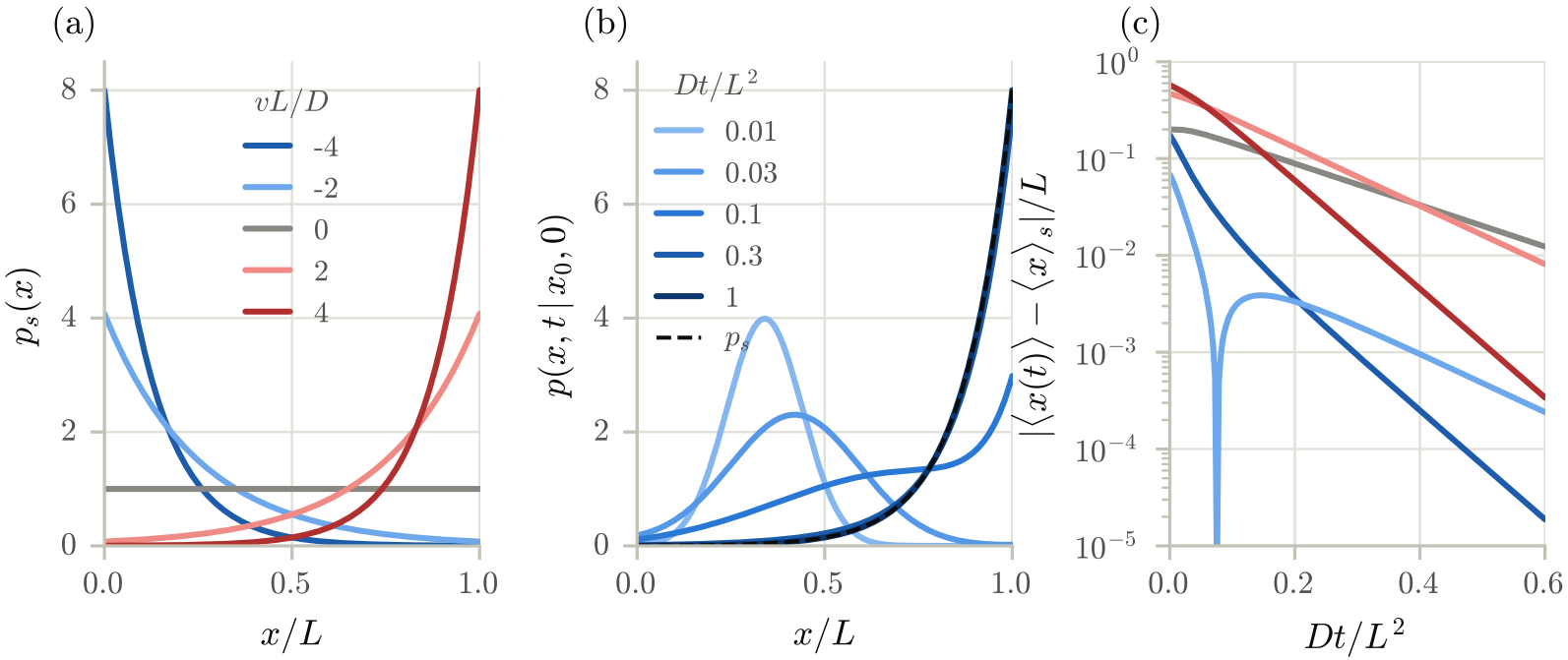
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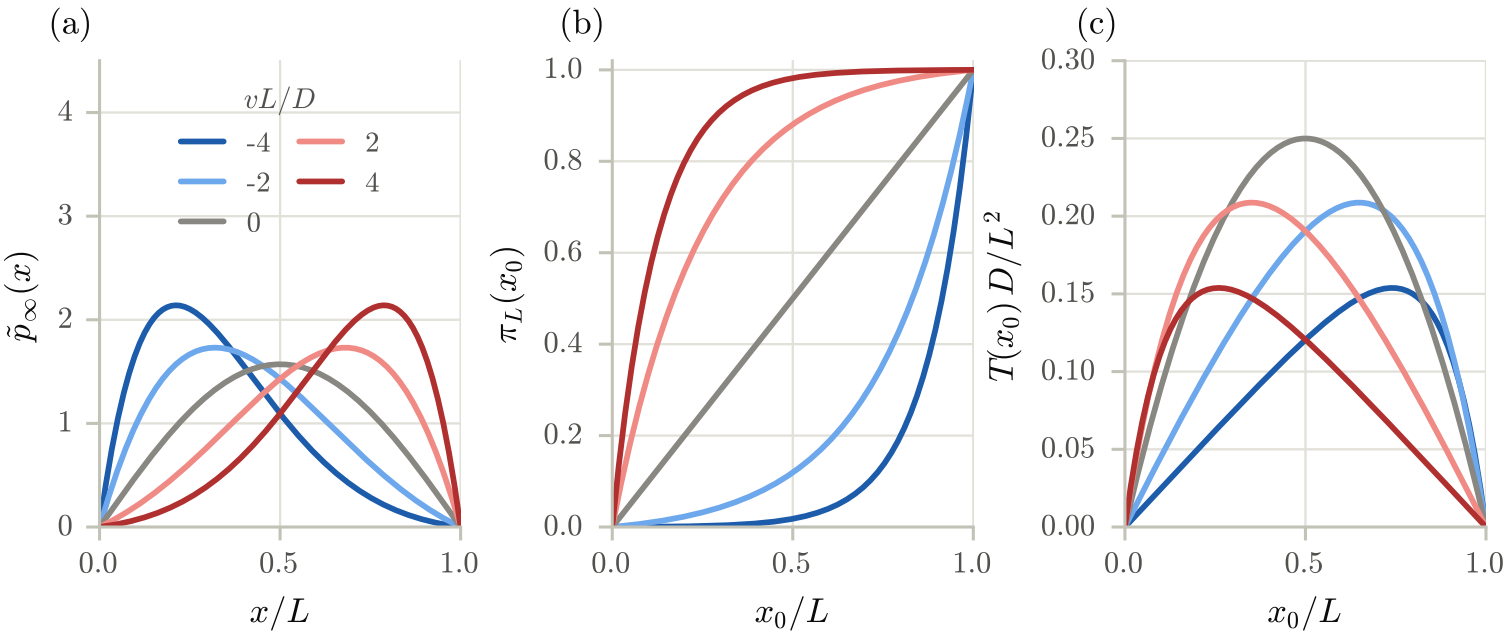
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- (b) $vL/D = 4$: carried downstream. (c) $\langle x \rangle$ relaxes at λ_1, the same for $\pm v$

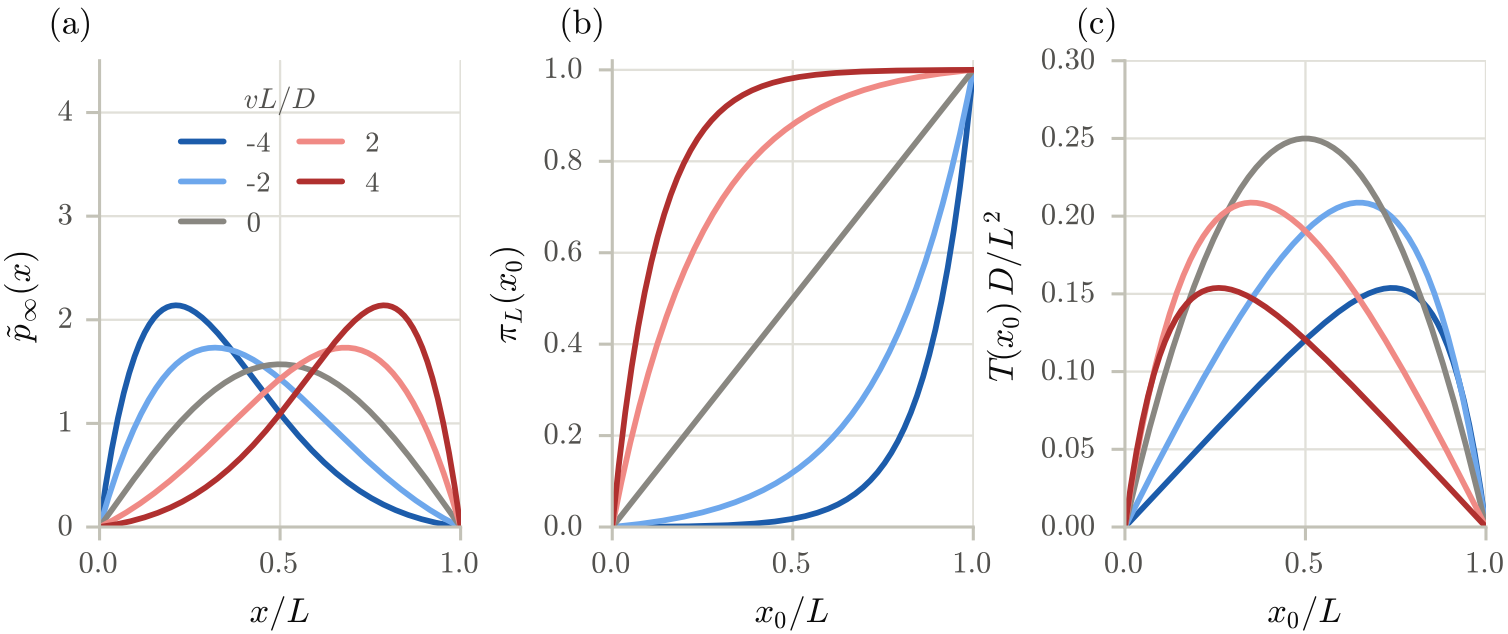
Constant Drift: Absorbing Walls

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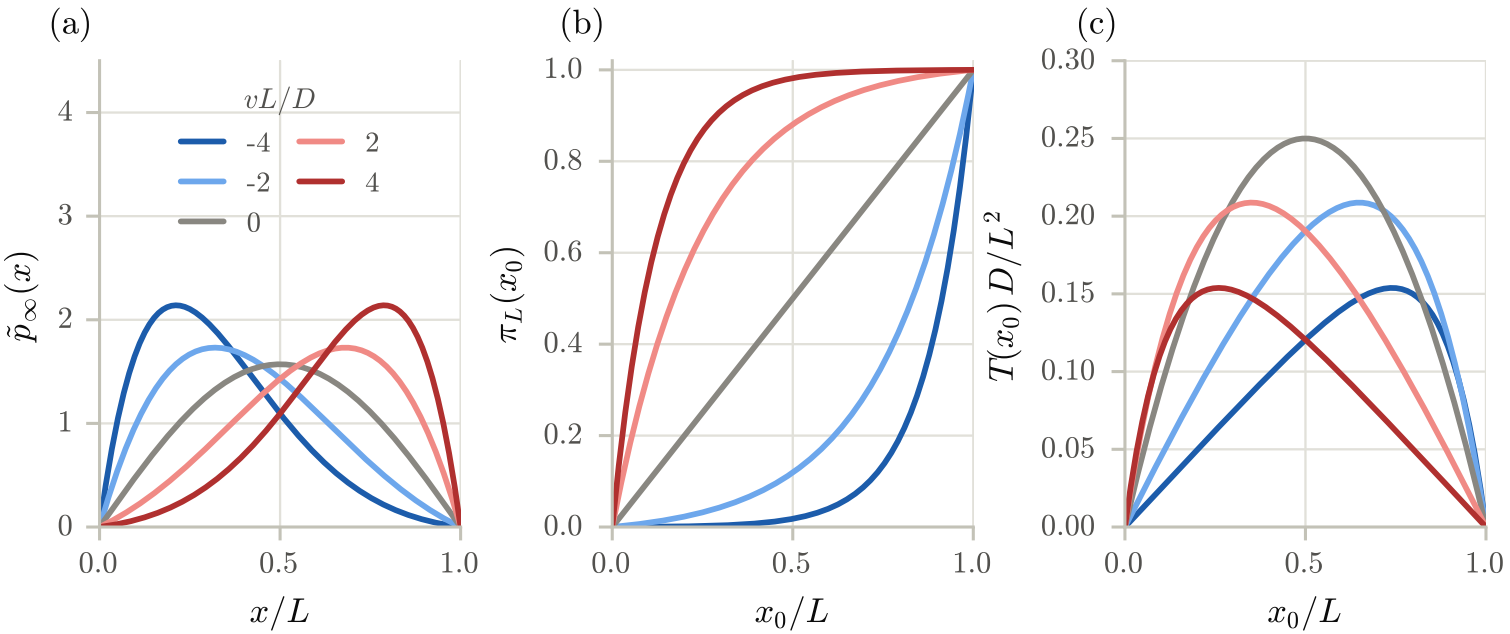
▸ Plot Lab ▸ SDE Explorer



- Tilted free sines: $P_n \propto e^{\alpha x} \sin k_n x$, $Q_n \propto e^{-\alpha x} \sin k_n x$, $k_n = n\pi/L$

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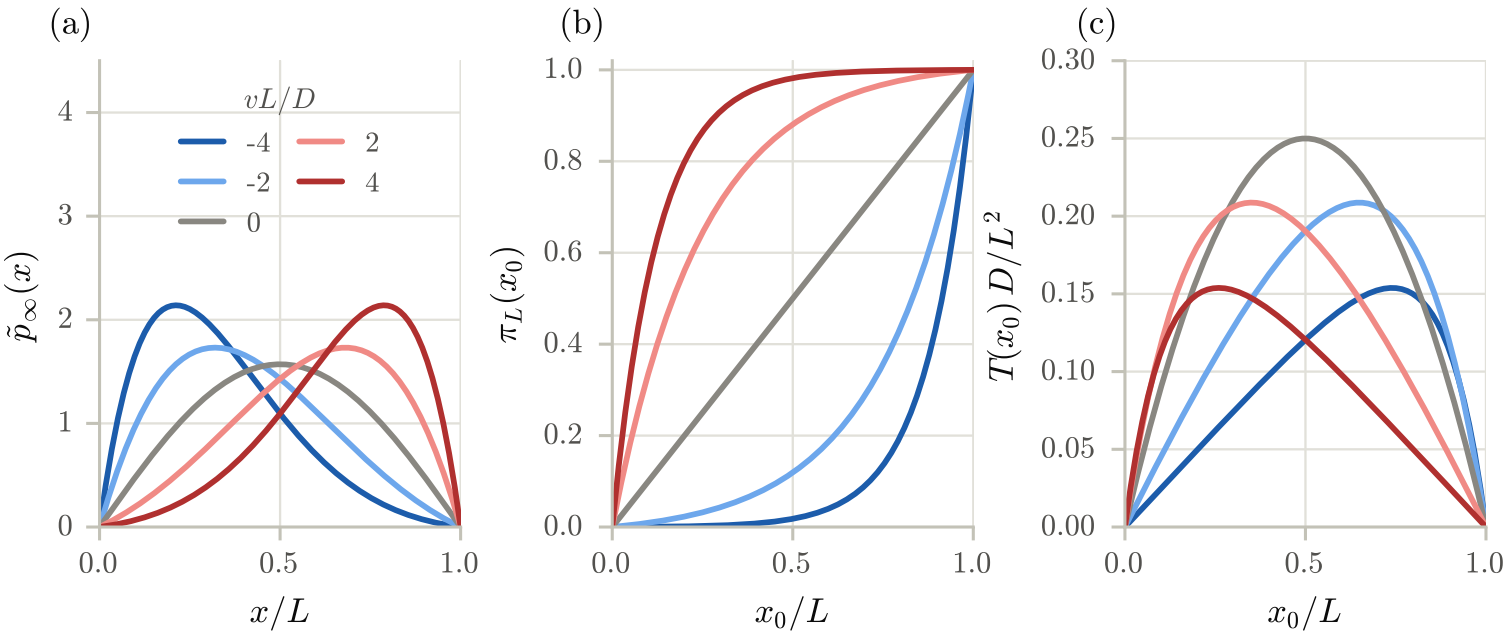
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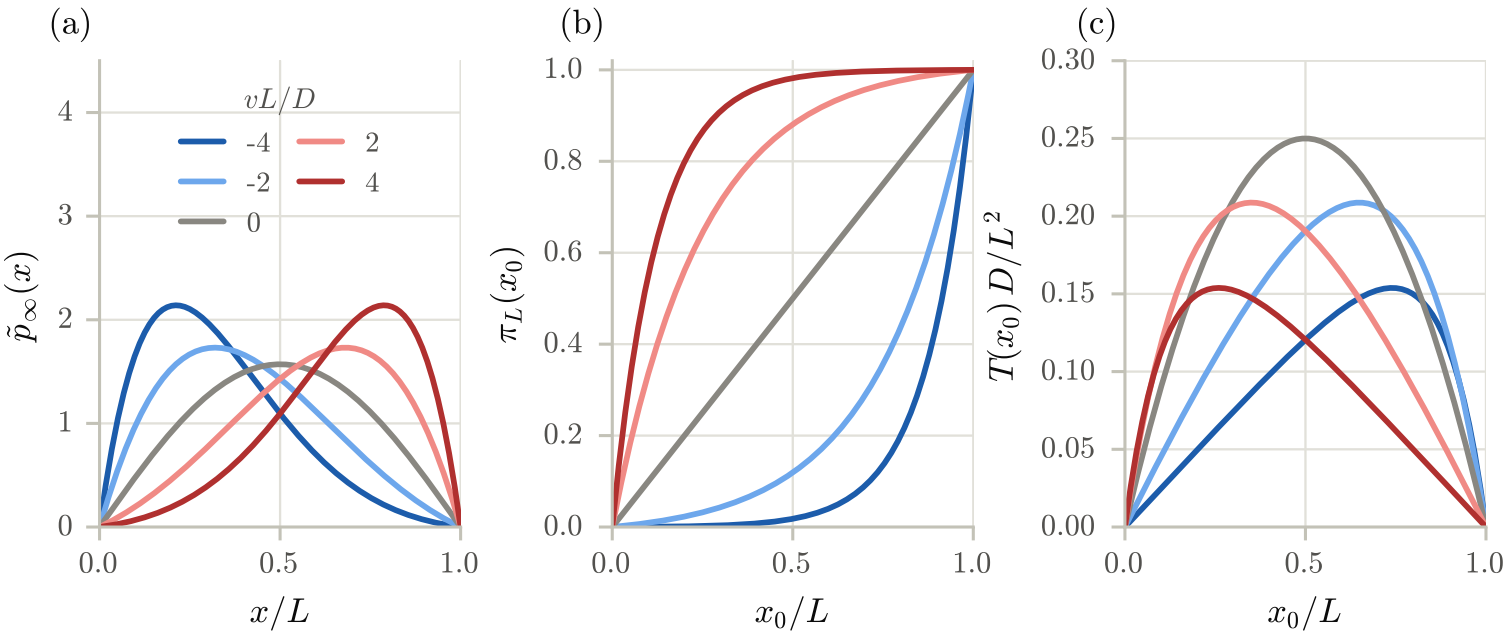
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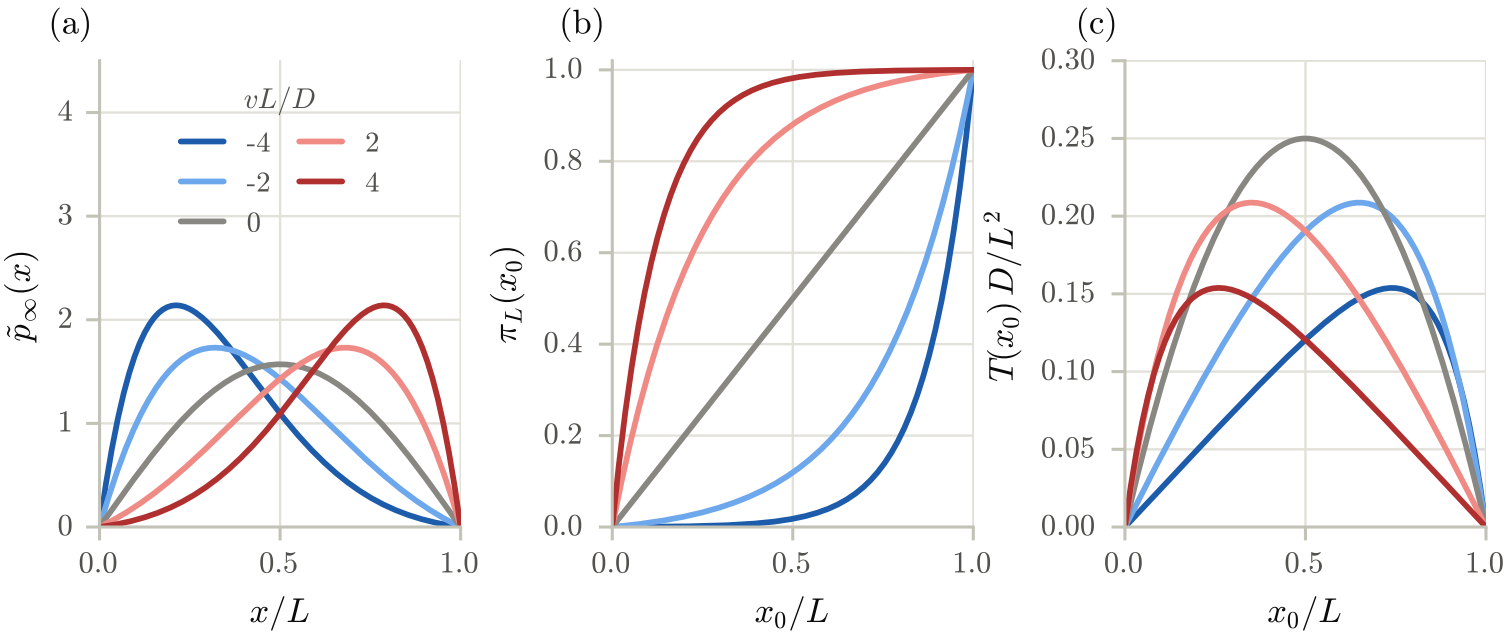
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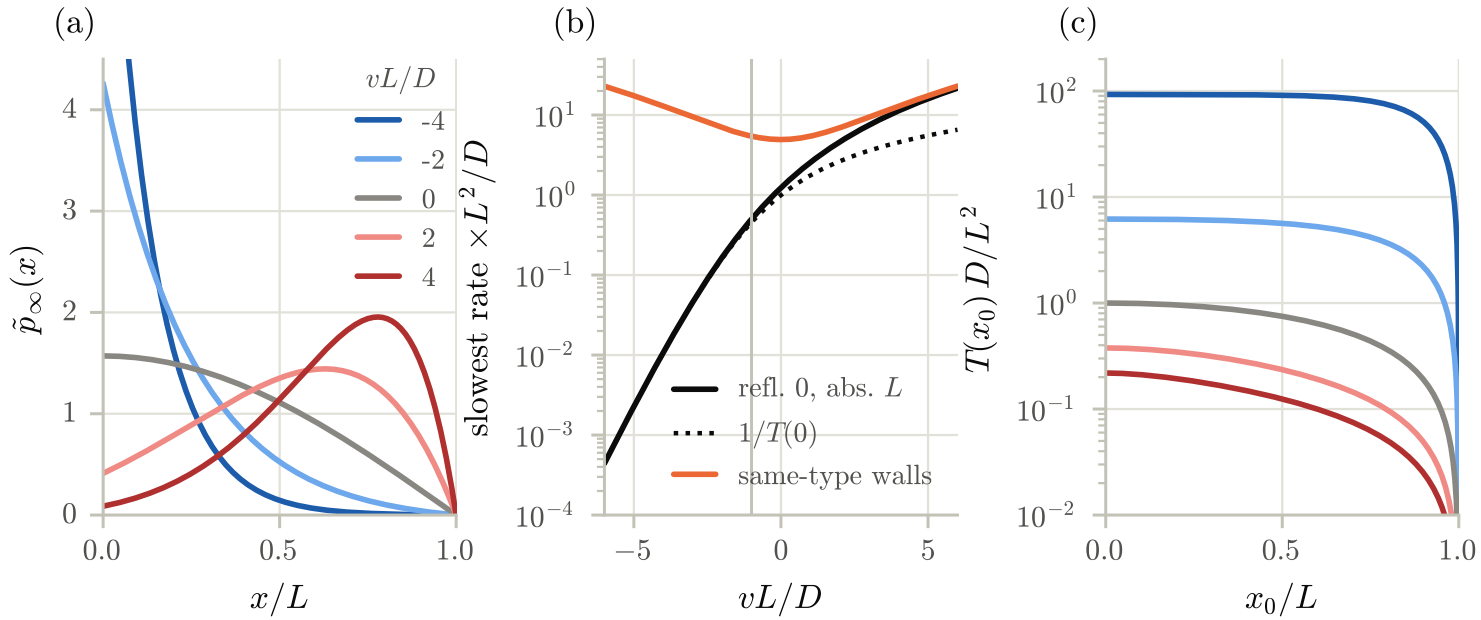
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- Mirror symmetry $x \to L - x$, $v \to -v$ pairs up the curves

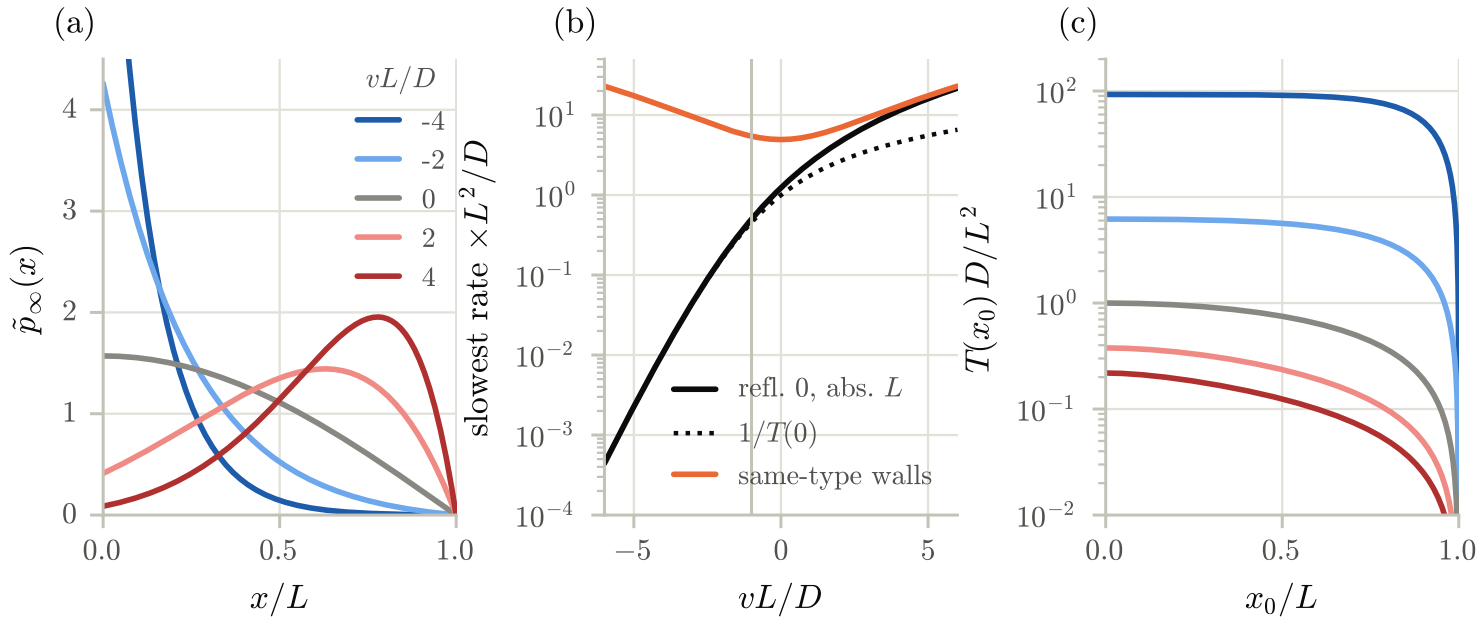
Constant Drift: Escape over a Barrier

▸ Plot Lab ▸ SDE Explorer



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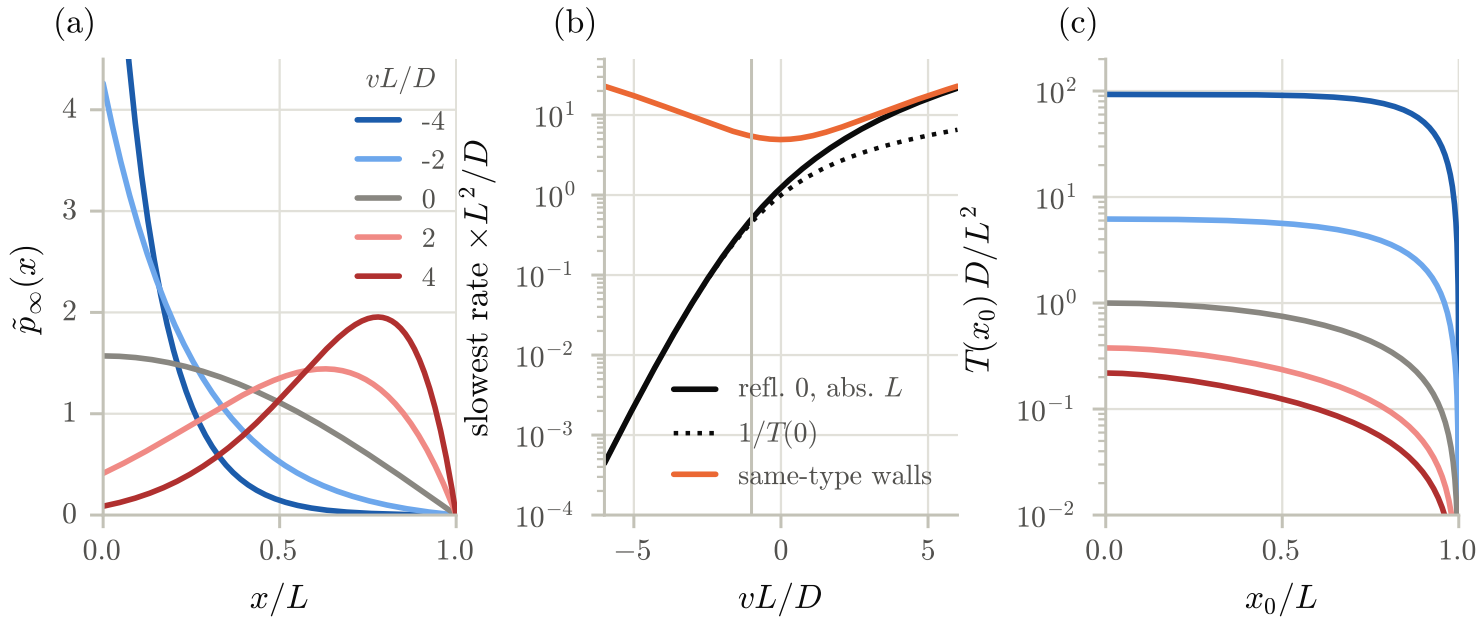
▸ Plot Lab ▸ SDE Explorer



- Reflecting at 0, absorbing at L: $\tan kL = -k/\alpha$

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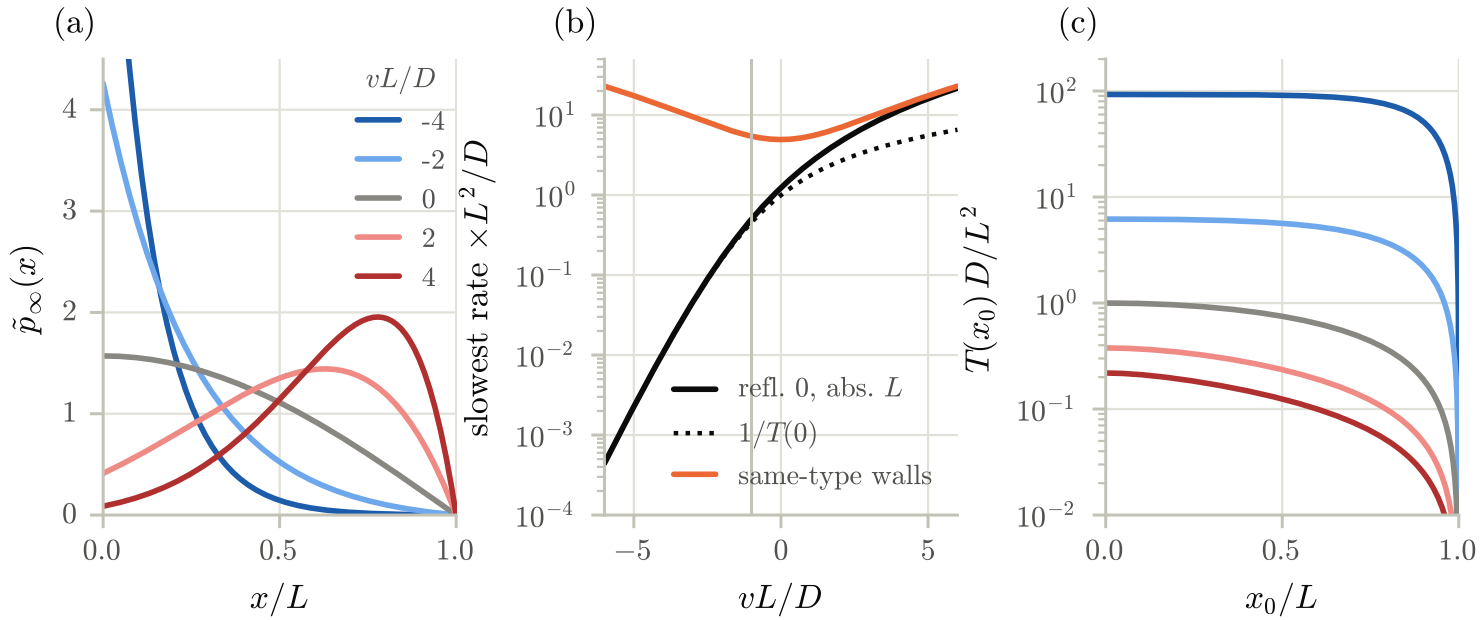
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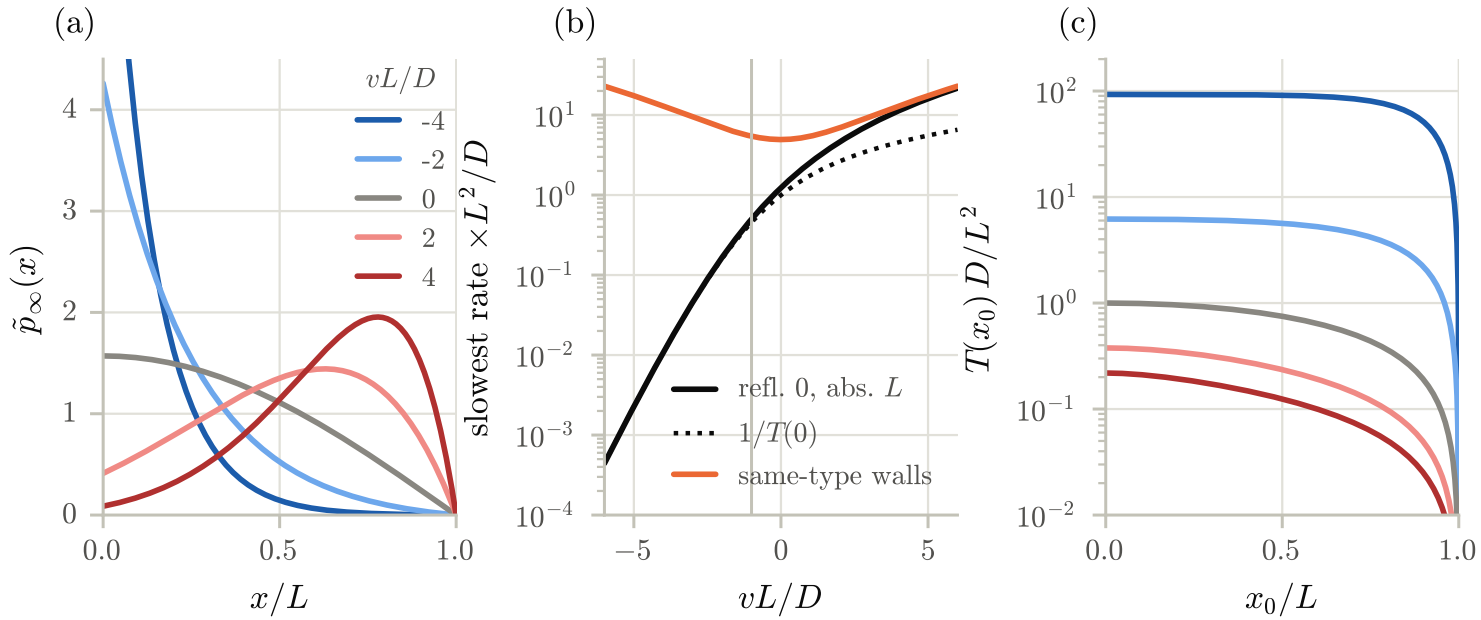
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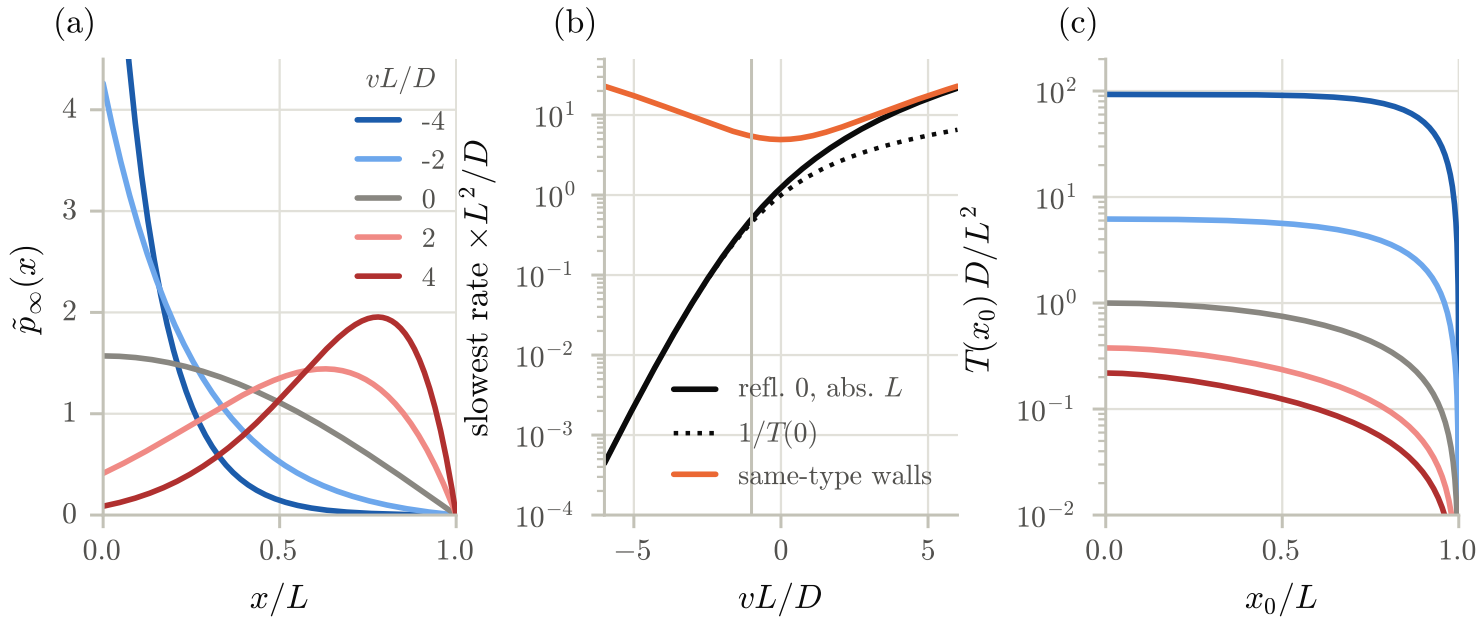
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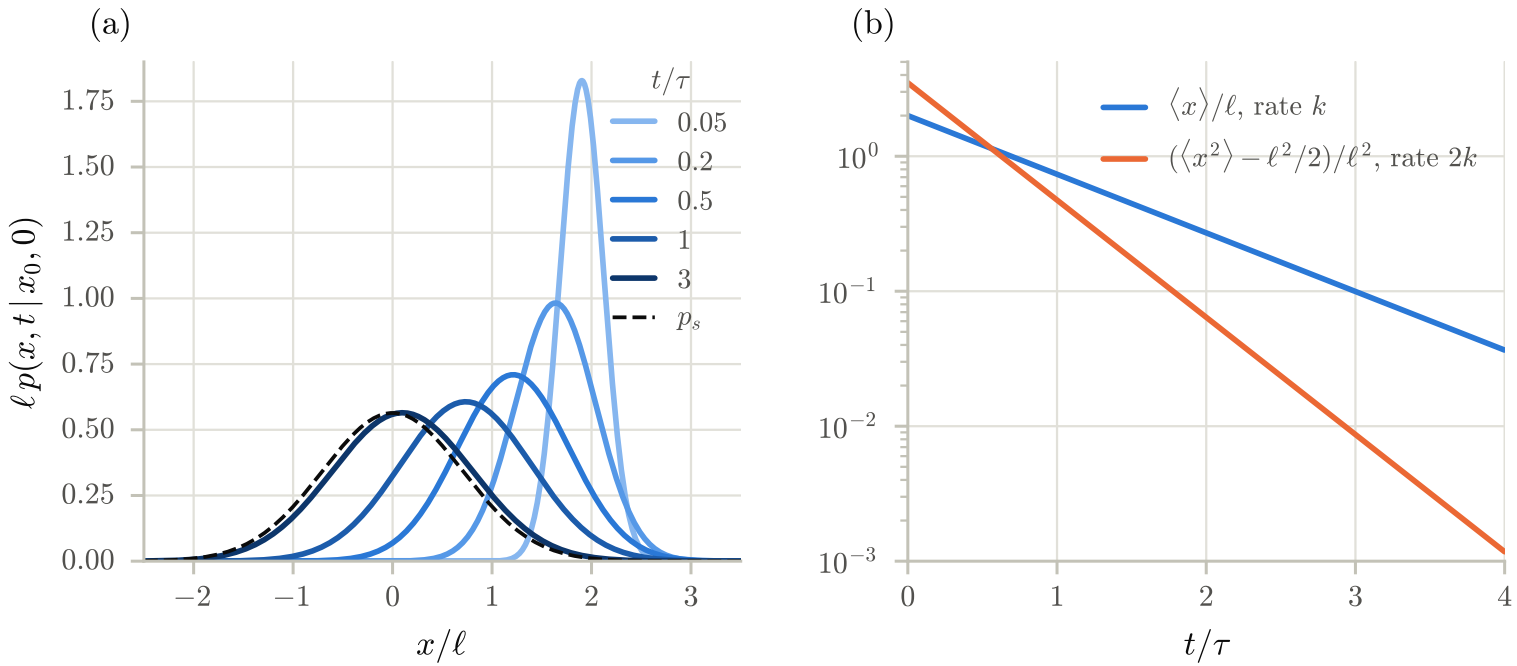
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- (c) $T(x_0)$ is flat except within $\sim \ell$ of the rim

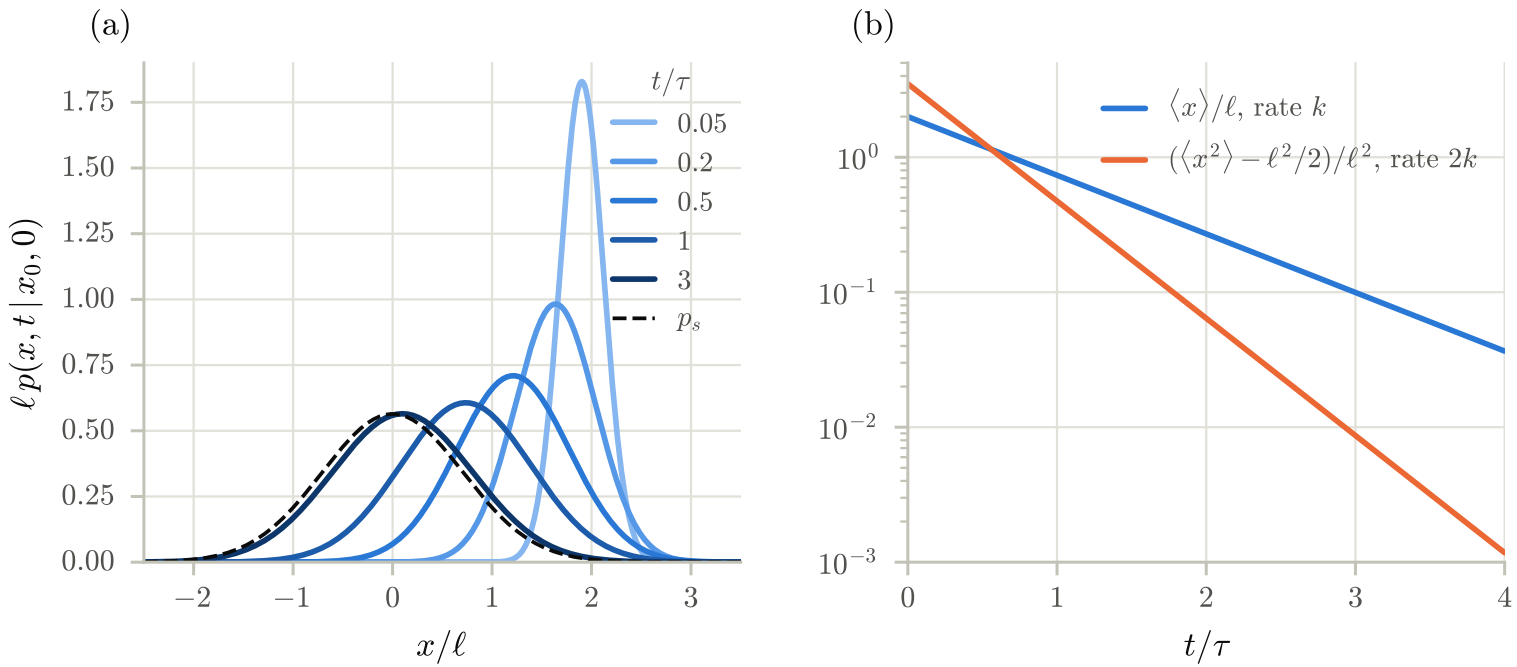
Example: Ornstein–Uhlenbeck

▸ Plot Lab ▸ SDE Explorer



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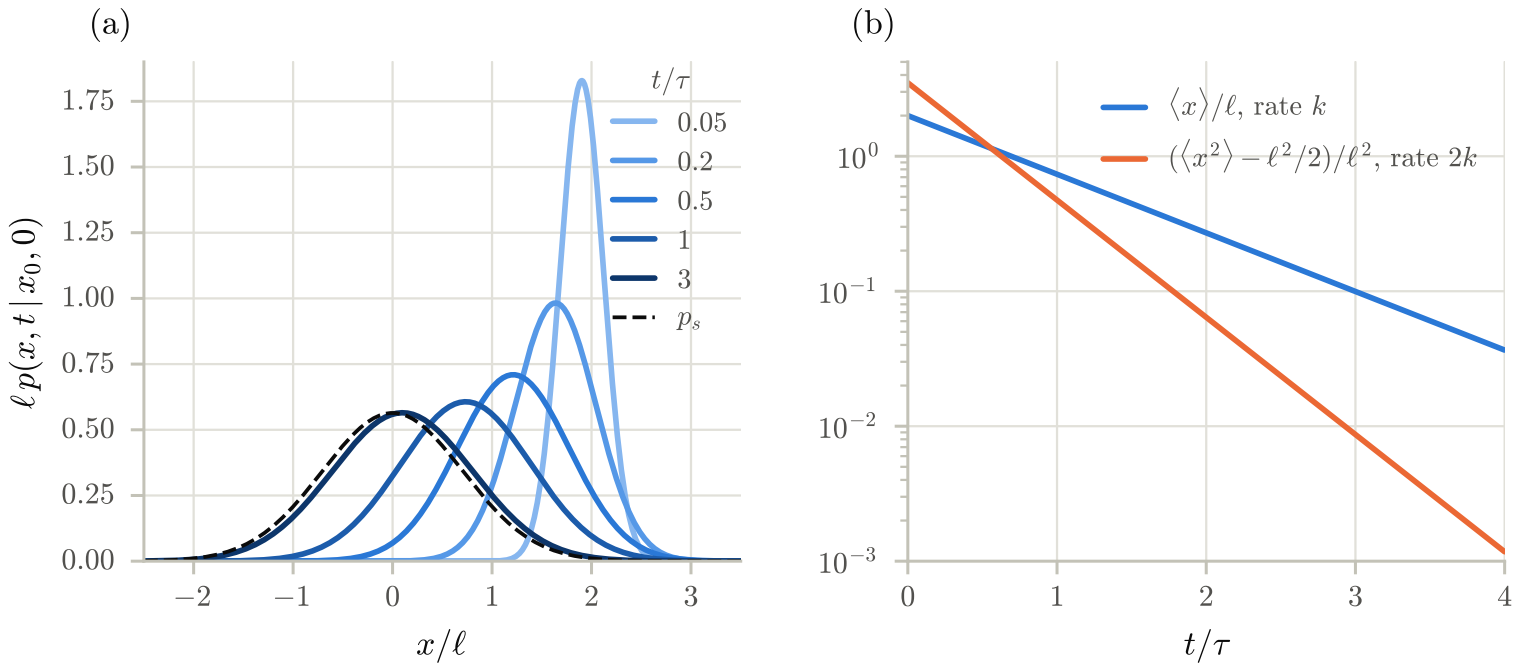
▸ Plot Lab ▸ SDE Explorer



- $A = -kx$, $B = D$: an optical tweezer ($k = \mathcal{K}/\gamma$, $\ell = \sqrt{D/k}$). Map $y = xe^{kt}$: free diffusion on the clock $\theta = (e^{2kt} - 1)/2k$

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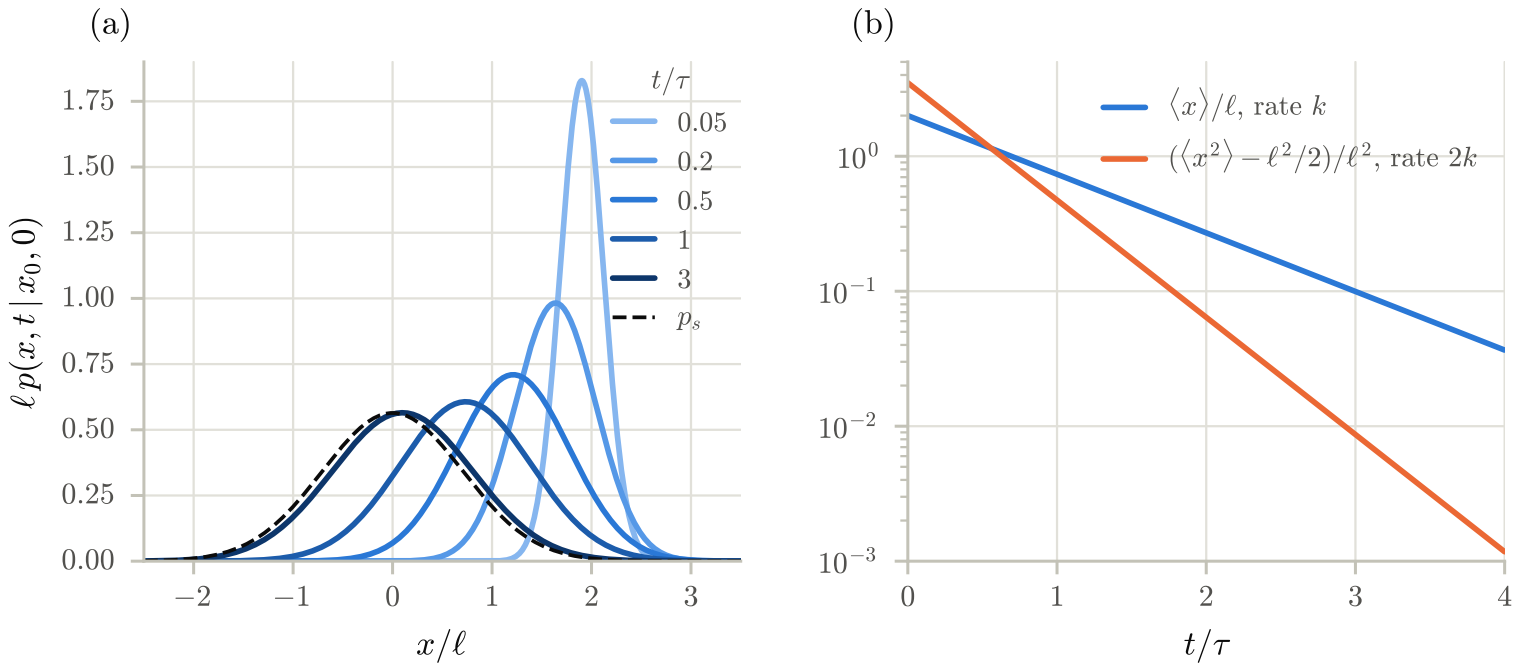
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- (b) Symmetry selects: $\langle x\rangle$ relaxes at k, but $\langle x^2\rangle$ at $2k$

Backup Slides

More examples

- Free diffusion on the line
- Walls by images
- Free diffusion on a ring
- Constant drift on the half-line
- Escape from a harmonic trap

Example: Free Diffusion on the Line

The root of the tree

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The root of the tree

- $A = 0$, $B = D$: a colloid in a bulk fluid

$$p(x, t \,|\, x_0, 0) = g(x - x_0, t), \qquad g(y, t) = \frac{1}{\sqrt{2\pi D t}}\, e^{-y^2/2Dt}$$

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- Translations: p depends only on $x - x_0$

Example: Free Diffusion on the Line

The root of the tree

- $A = 0$, $B = D$: a colloid in a bulk fluid

$$p(x, t \,|\, x_0, 0) = g(x - x_0, t), \qquad g(y, t) = \frac{1}{\sqrt{2\pi D t}}\, e^{-y^2/2Dt}$$

- Modes e^{ikx}, $\lambda = \frac{1}{2} D k^2$: a continuum down to 0
  - No gap and no stationary state: algebraic decay, $p(x_0, t) \propto t^{-1/2}$
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Example: Free Diffusion on the Line

The root of the tree

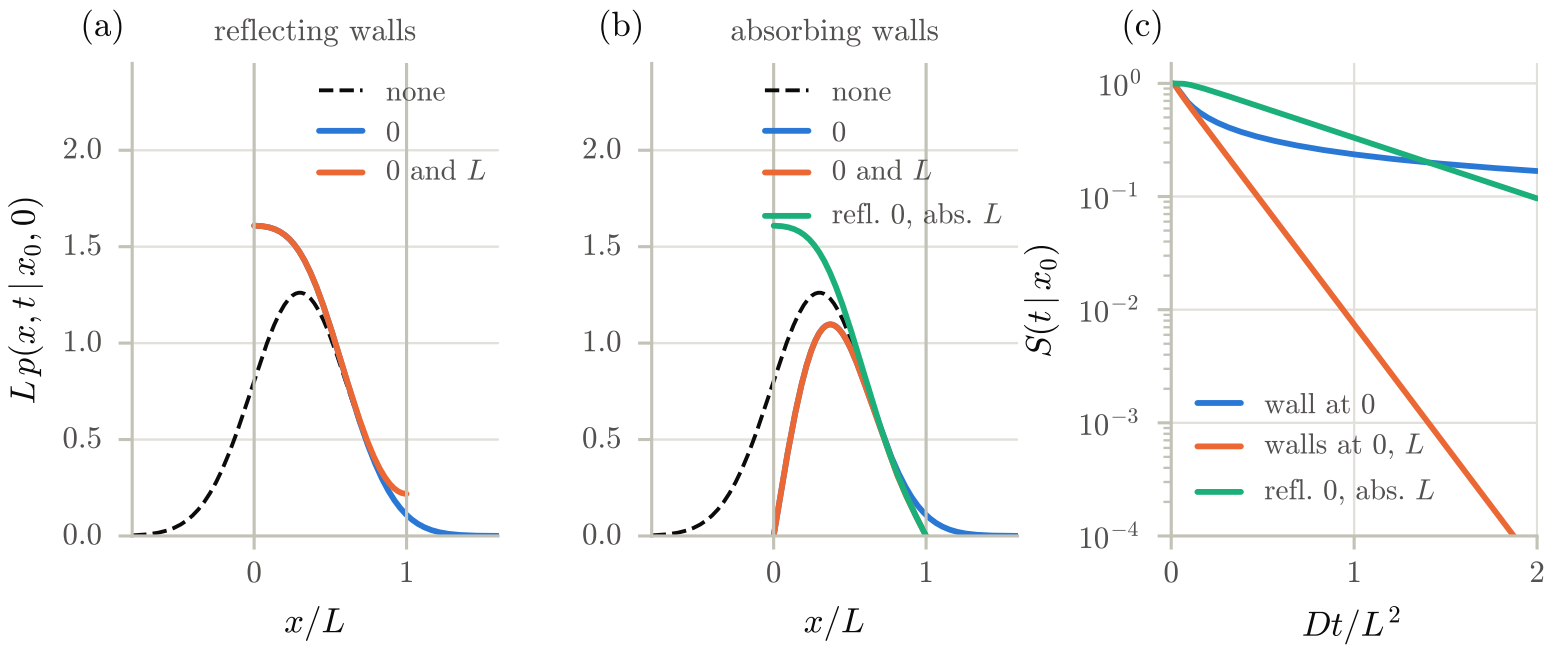
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- Everything below is built from g

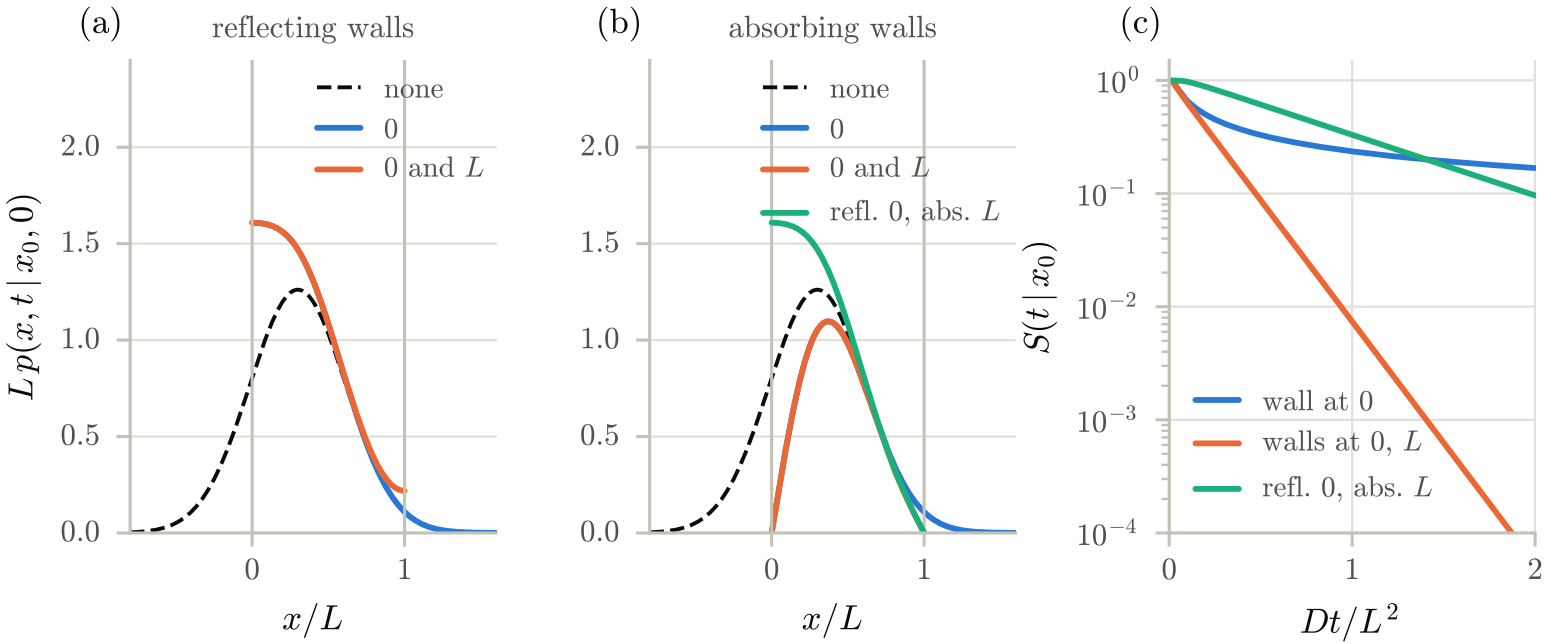
Example: Walls by Images

▸ Plot Lab



Example: Walls by Images

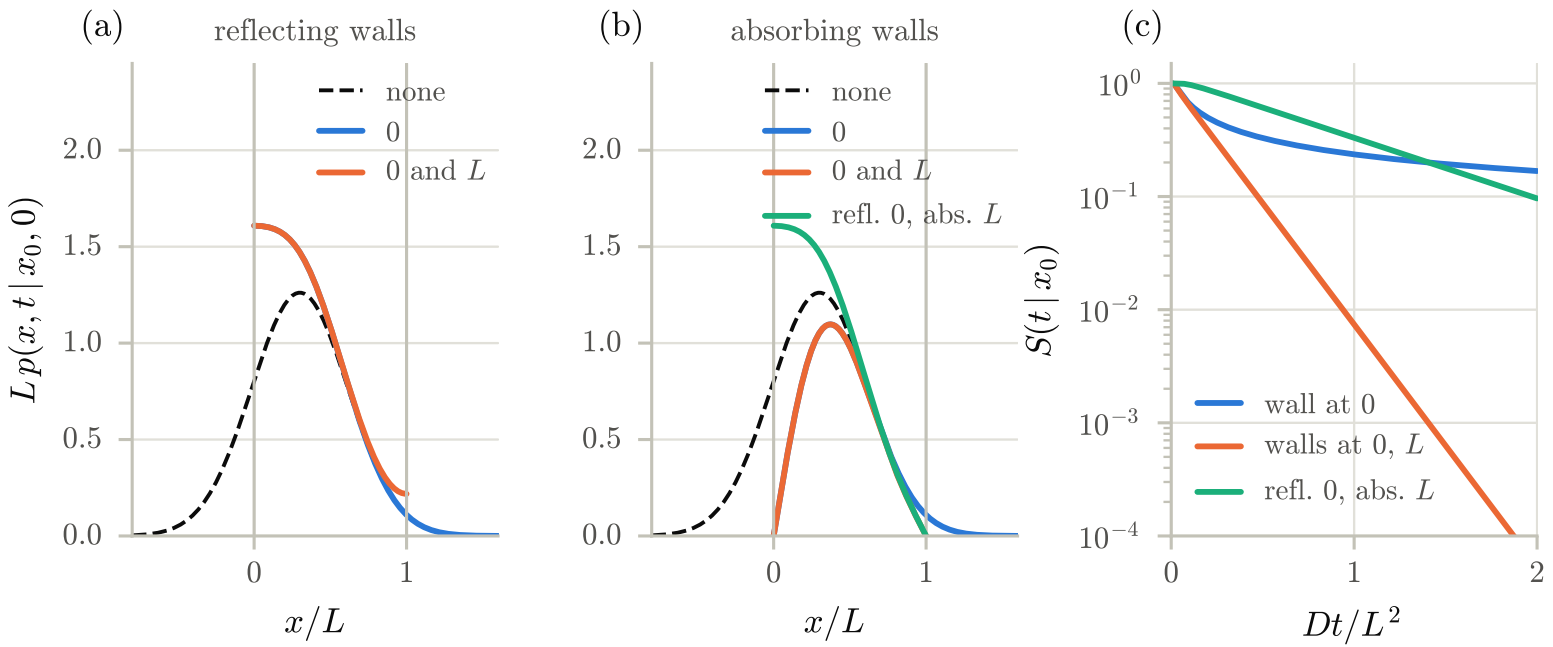
▸ Plot Lab



- Wall at 0: add a mirror image at $-x_0$, $g(x-x_0) \pm g(x+x_0)$; $+$ is even ($J=0$), $-$ is odd ($p=0$)

Example: Walls by Images

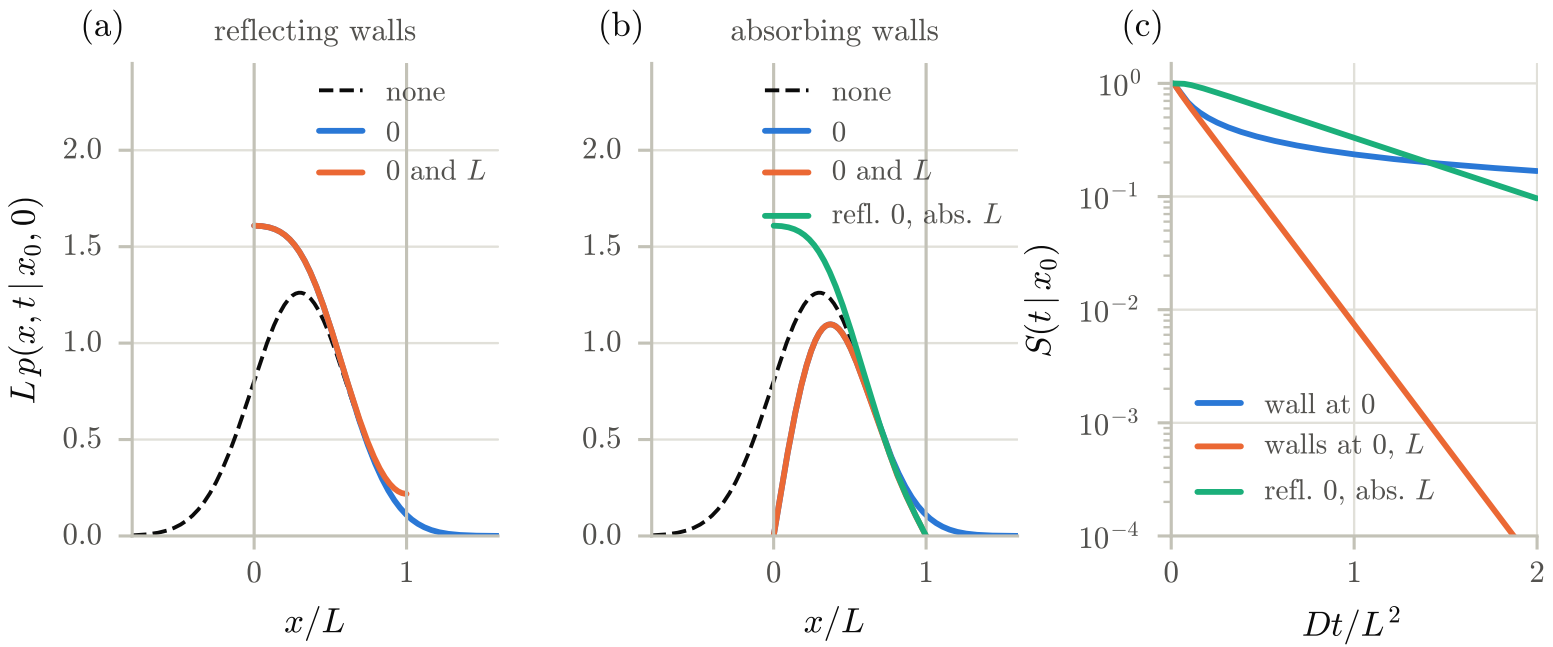
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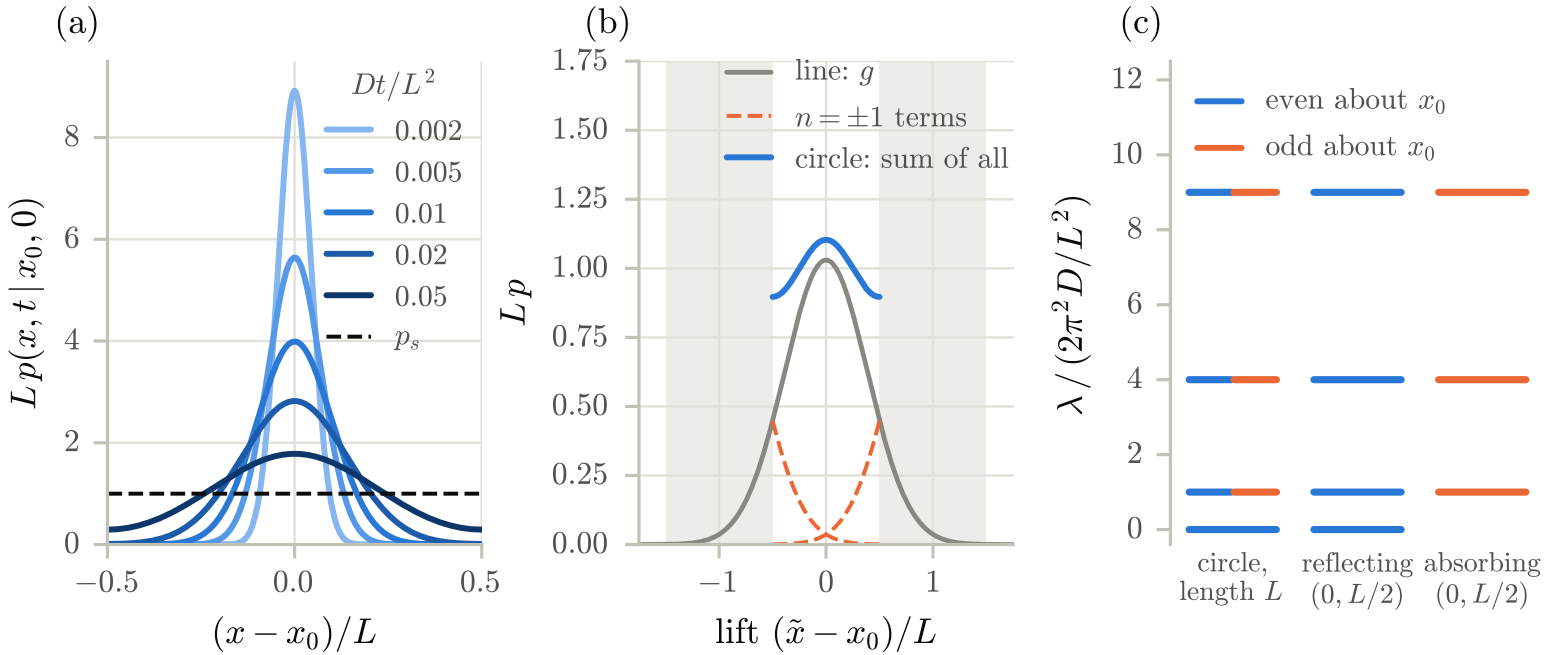
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- Two walls: images of images, an infinite sum $\iff$ the $\cos$ (reflecting) or $\sin$ (absorbing) modes

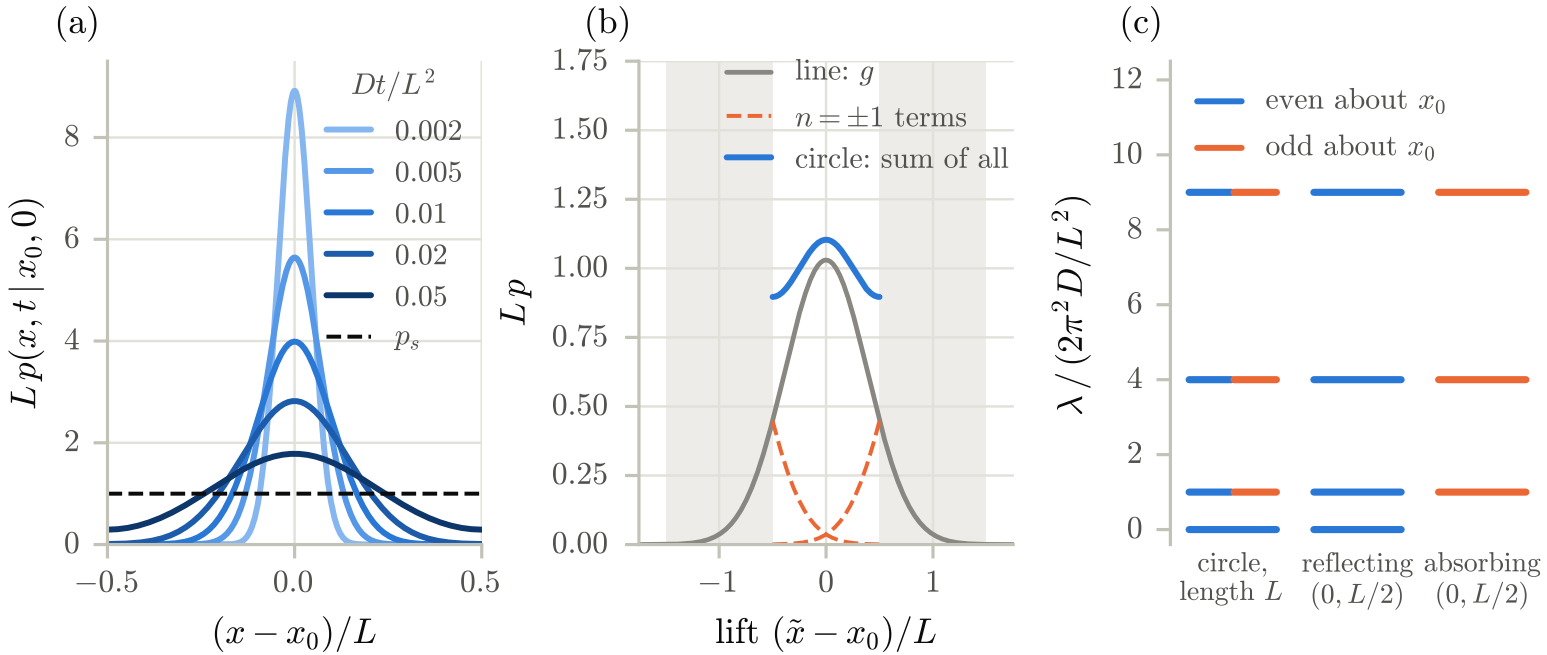
Example: Free Diffusion on a Ring

▸ Plot Lab ▸ SDE Explorer



Example: Free Diffusion on a Ring

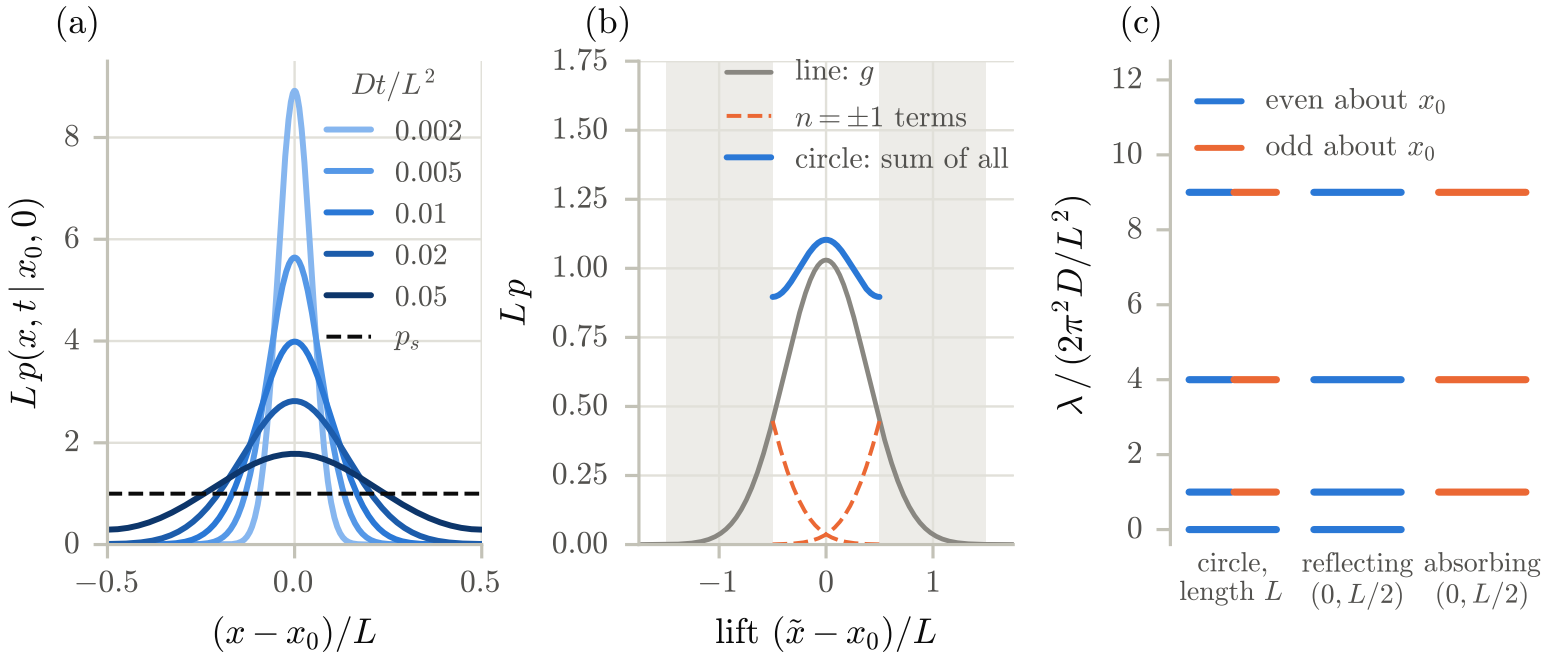
▸ Plot Lab ▸ SDE Explorer



- (a) Relaxes to $p_s = 1/L$; modes $e^{\pm i q_m x}$, $\lambda_m = \frac{1}{2} D q_m^2$, $q_m = 2\pi m/L$

Example: Free Diffusion on a Ring

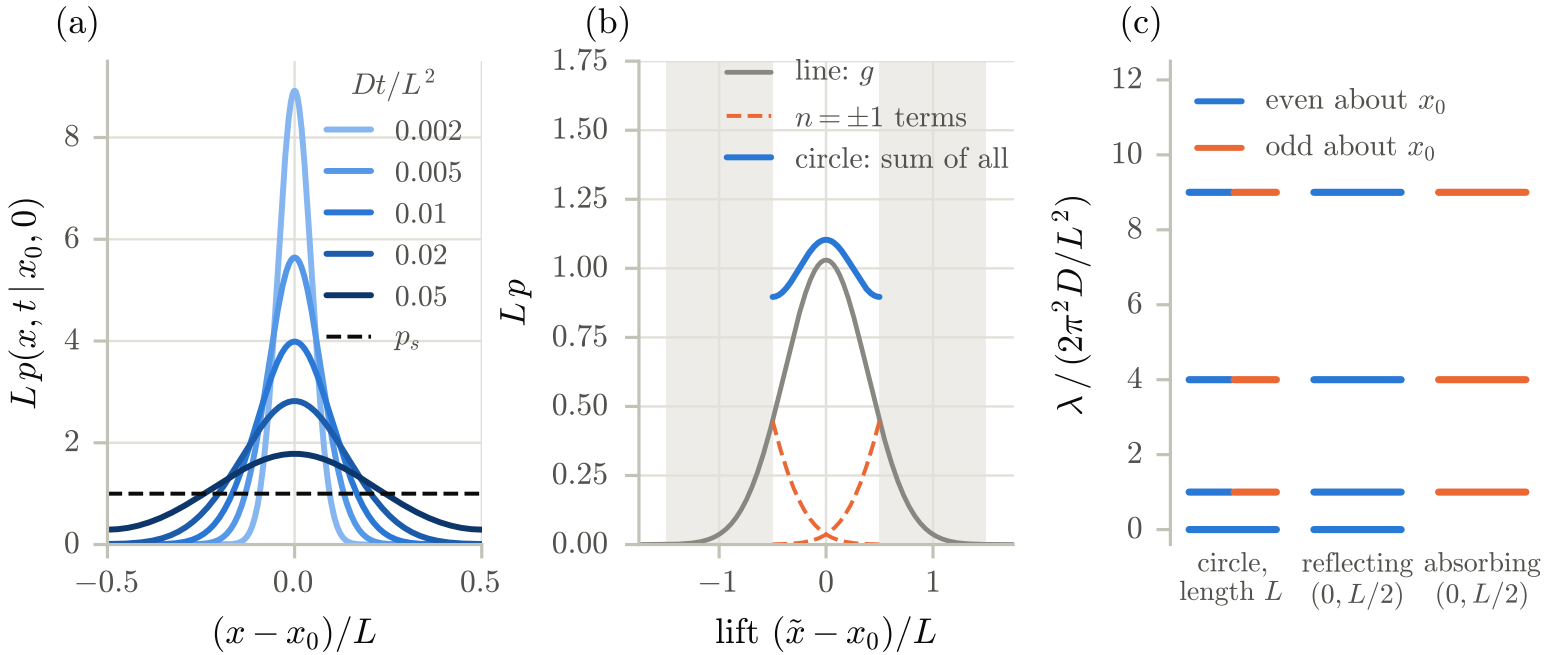
▸ Plot Lab ▸ SDE Explorer



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- (b) The cover: the line's Gaussian plus all its translates by multiples of L

Example: Free Diffusion on a Ring

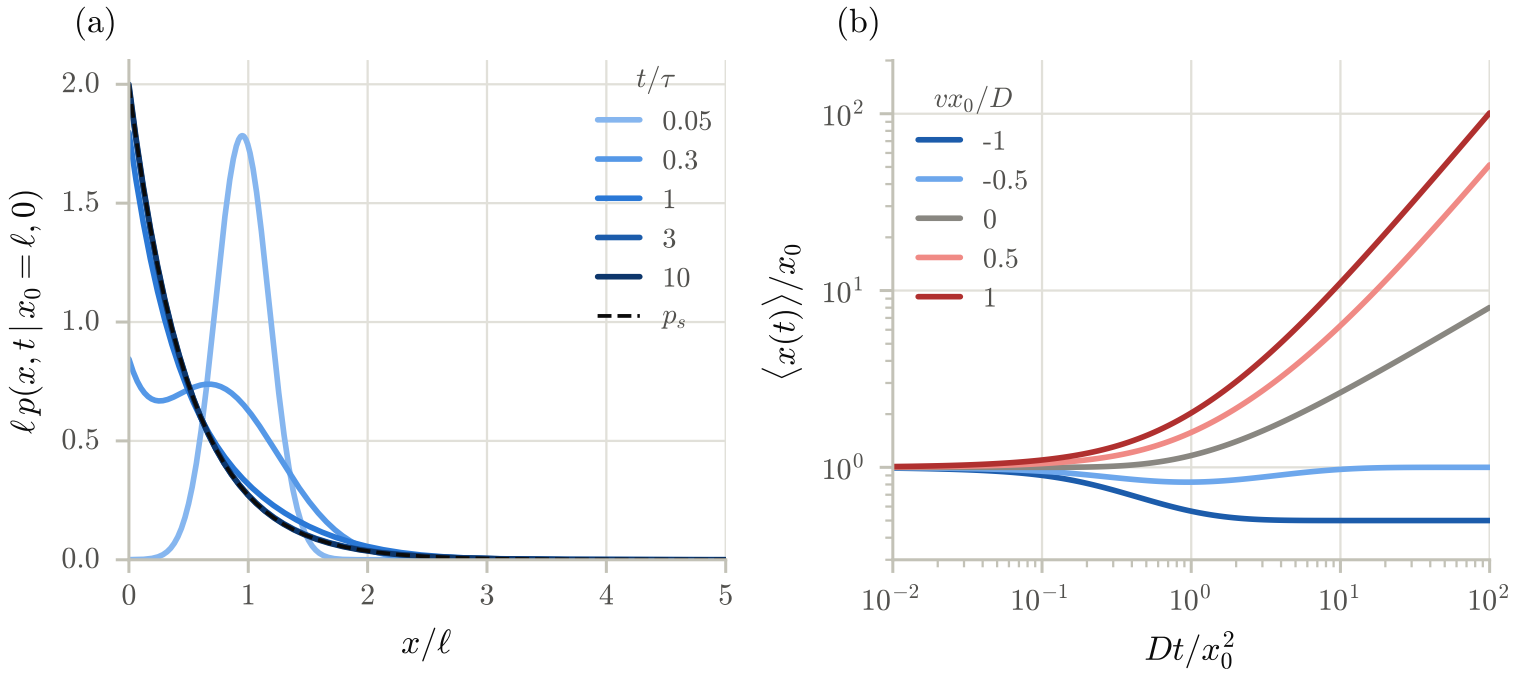
▸ Plot Lab ▸ SDE Explorer



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- (b) The cover: the line's Gaussian plus all its translates by multiples of L
- (c) Folding about x_0: even modes are the reflecting interval $(0, L/2)$, odd ones the absorbing interval

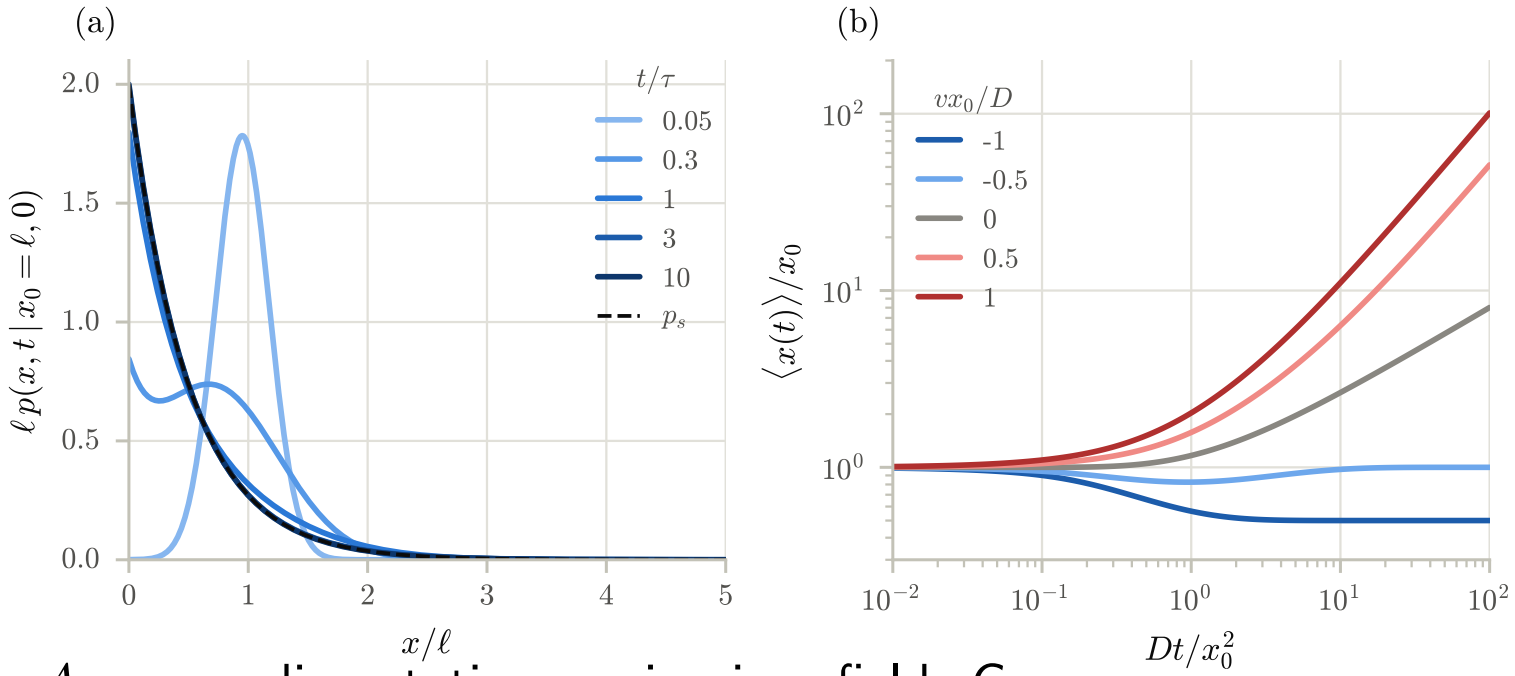
Example: Constant Drift on the Half-Line

▸ Plot Lab ▸ SDE Explorer



Example: Constant Drift on the Half-Line

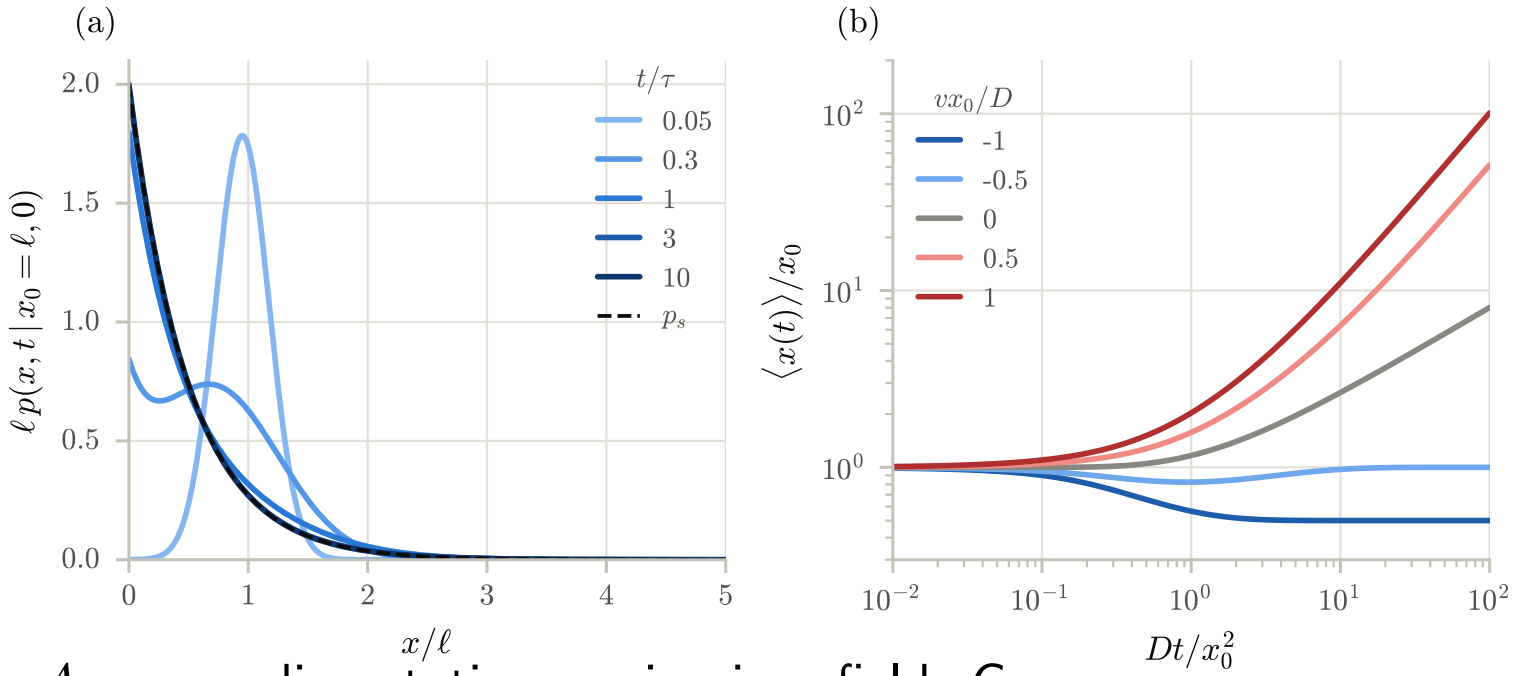
▸ Plot Lab ▸ SDE Explorer



- $A = v$: sedimentation, an ion in a field. Gauge: $p = e^{\alpha(x-x_0)-t/2\tau}\, u$, $\alpha = v/D$, $\tau = D/v^2$, u free

Example: Constant Drift on the Half-Line

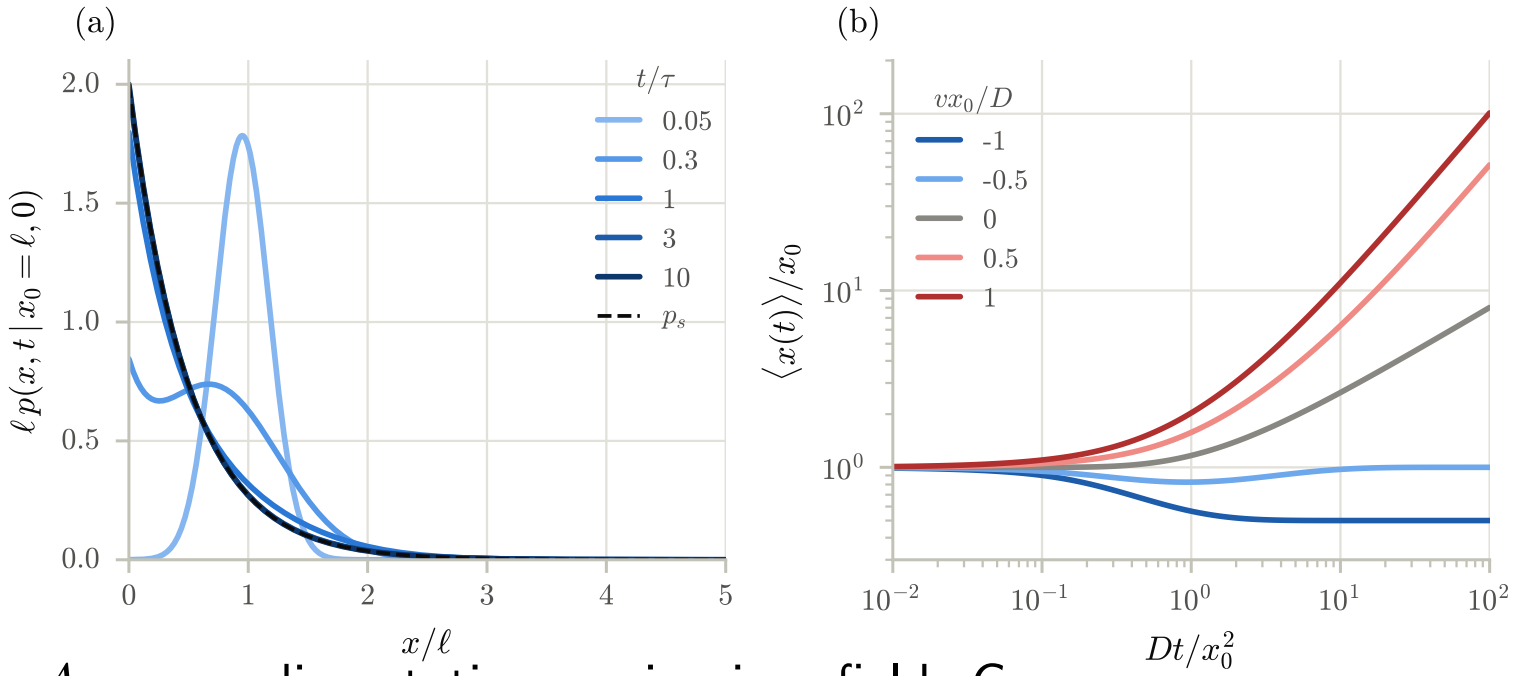
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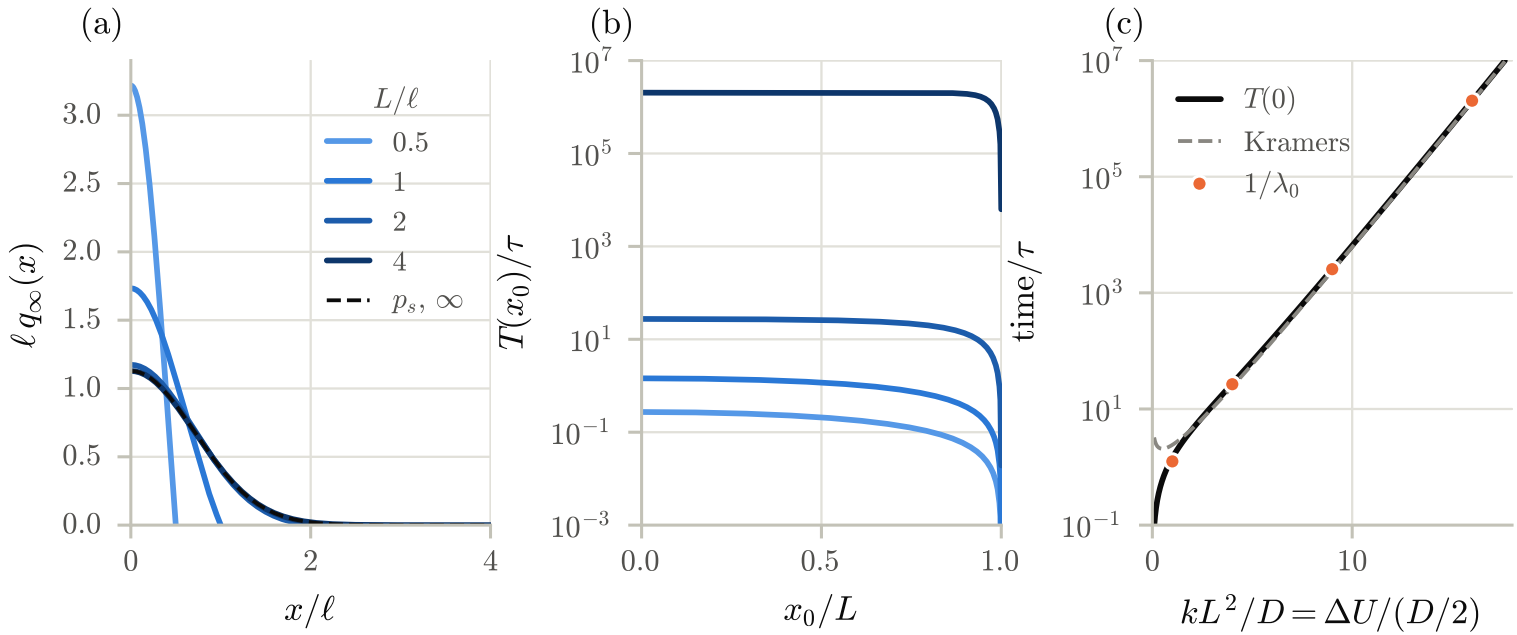
▸ Plot Lab ▸ SDE Explorer



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- Rates are those of u plus $1/2\tau$. Absorbing walls stay absorbing; reflecting ones become Robin, $u' = \alpha u$
- (a) Floor with $v < 0$: barometric $p_s \propto e^{-2|v|x/D}$; spectrum $\{0\} \cup [1/2\tau, \infty)$. (b) $\langle x(t)\rangle$

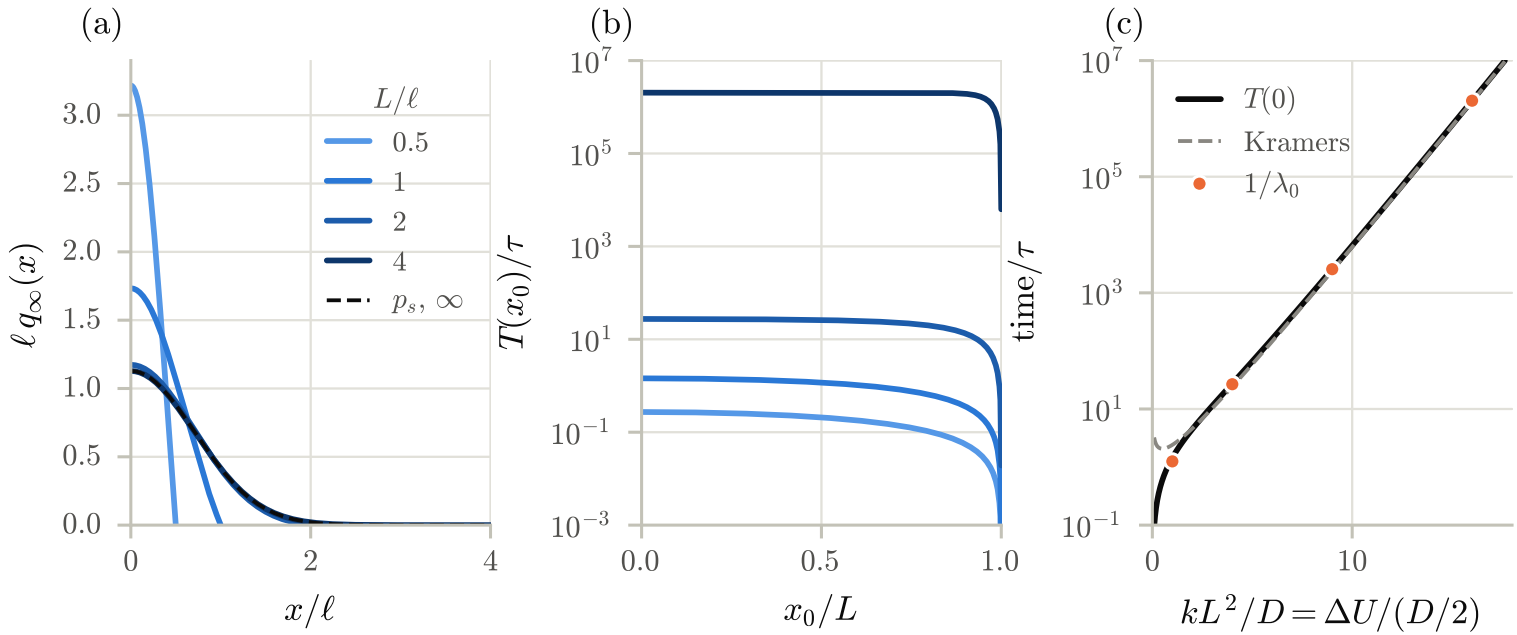
Example: Escape from a Trap

▸ Plot Lab ▸ SDE Explorer



Example: Escape from a Trap

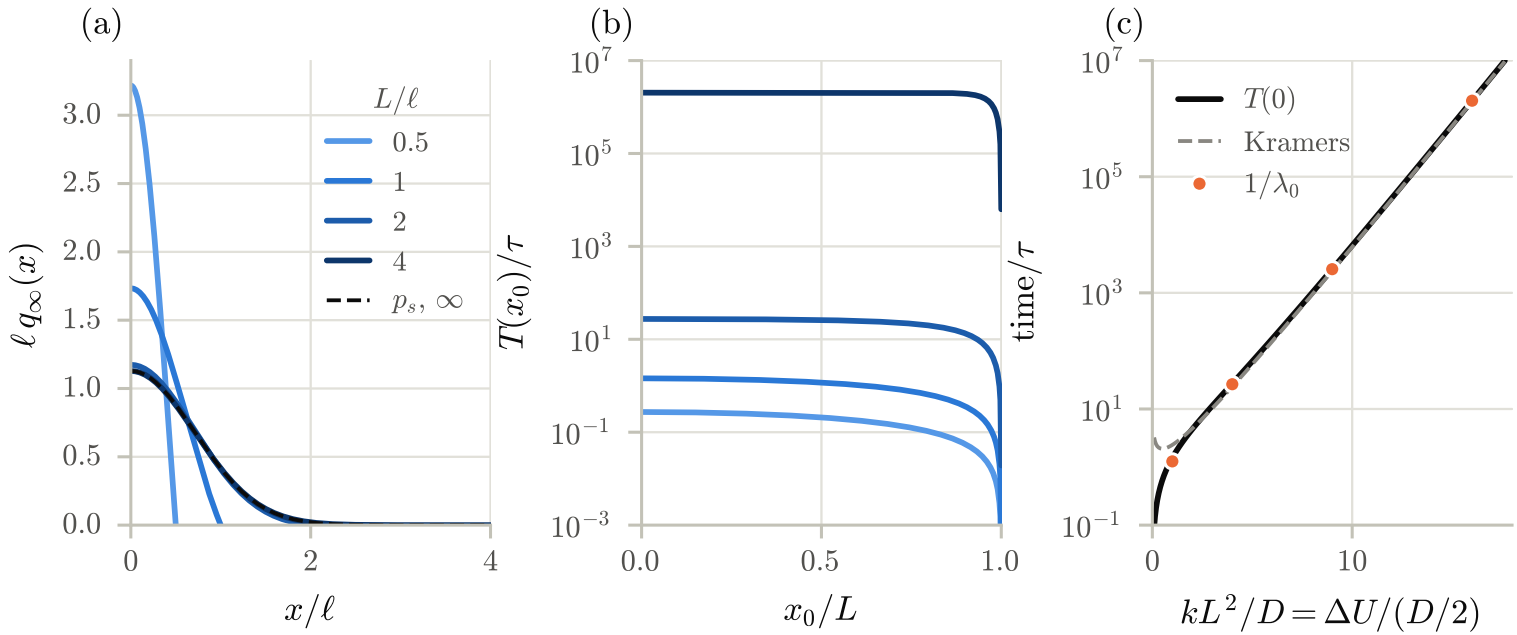
▸ Plot Lab ▸ SDE Explorer



- OU, reflecting at 0, absorbing at L. The map moves this wall: Kummer functions, no further reduction

Example: Escape from a Trap

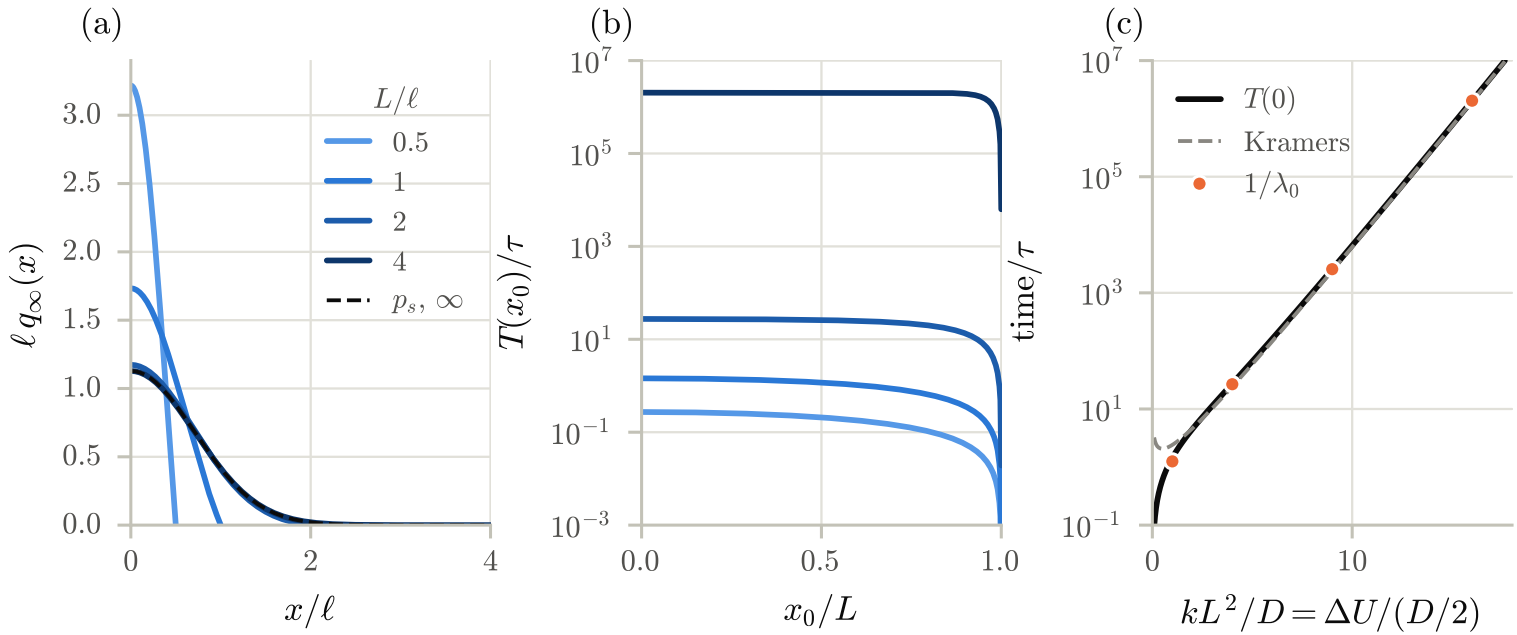
▸ Plot Lab ▸ SDE Explorer



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- (a) QSD $\to$ half-Gaussian: equilibrated in the well. (b) $T(x_0)$ is flat except near the exit

Example: Escape from a Trap

▸ Plot Lab ▸ SDE Explorer



- OU, reflecting at 0, absorbing at L. The map moves this wall: Kummer functions, no further reduction
- (a) QSD $\to$ half-Gaussian: equilibrated in the well. (b) $T(x_0)$ is flat except near the exit
- (c) $\lambda_0 \simeq 1/T \simeq \frac{2k}{\sqrt{\pi}} \frac{L}{\ell}\, e^{-L^2/\ell^2}$ (Kramers); the next level is $\sim 2k$: metastability